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High-precision gravity measurements using atom interferometry

A. Peters, K. Y. Chung and S. Chu

Abstract. We have built an atom interferometer that can measure g, the local acceleration due to gravity, with a resolution of $\Delta g/g = 2\times10^{-8}$ after a single $1.3\,\mathrm{s}$ measurement cycle, 3×10^{-9} after 1 min and 1×10^{-10} after two days of integration time. The difference between our value for g and one obtained by a falling corner-cube optical interferometer is $(7\pm7)\times10^{-9}$ g. The atom interferometer uses velocity-selective stimulated Raman transitions and laser-cooled caesium atoms in an atomic fountain. We extend previous methods of analysing the interferometer to include the effects of a gravitational gradient. We also present detailed experimental and theoretical studies of potential systematic errors and noise sources.

1. Introduction

Matter-wave interferometers using neutral atoms were first demonstrated in 1991. Of the four atom interferometers realized that year, two used diffraction by micro-fabricated slits [1] or gratings [2]. The other two interferometers made use of the momentum recoil associated with the absorption or stimulated emission of optical photons during one-photon [3] or two-photon transitions [4]. Subsequent experiments have also shown the feasibility of using diffraction by periodic light fields [5]. More recent atom interferometer work is presented in a book edited by Berman [6].

Since 1991 the performance of atom interferometers has improved substantially. Recently, they have been used to measure rotations [7] and fundamental constants [8, 9] with a performance matching that of any competing technique. We have used our atom interferometer to measure g, the acceleration due to gravity. A summary of our results can be found in [10]. This paper describes in detail the theoretical and experimental aspects of our measurements.

Measuring g, a site-dependent quantity that also varies with time, is interesting for a variety of reasons, including applications in geophysics and metrology [11]. Precise measurements of g are also useful for locating oil, water and other natural resources. These applications require measurement accuracies ranging from 10^{-6} g to 10^{-9} g. Accordingly, instruments which measure gravity come in many different forms and with many performance parameters (Table 1). Most of them provide only relative gravity readings. Currently, the best absolute gravimeters are falling corner-cube instruments which can obtain a gravity value accurate to approximately 2 μGal (1 μGal $= 10^{-8}$ $\mathrm{m/s^2} \approx 10^{-9}$ g). They are a proven reference against which we can compare our interferometer (Section 4.4).

The atom interferometer we use to measure g is an improved version of the one used by Kasevich and Chu in 1991 [4] and uses stimulated Raman transitions between hyperfine ground states to implement beam splitters and mirrors. Stimulated Raman transitions have been discussed elsewhere in great detail [16, 17]. The basic process is illustrated in Figure 1 and can be summarized as follows:

(a) The atom is illuminated by two counter-propagating laser beams whose frequency difference equals the splitting between two hyperfine ground states of the atom. We can assign them an effective wave number $\boldsymbol{k}_{\mathrm{eff}} = \boldsymbol{k}_1 - \boldsymbol{k}_2$ ($k_{\mathrm{eff}} = |\boldsymbol{k}_2| + |\boldsymbol{k}_1|$ for counter-

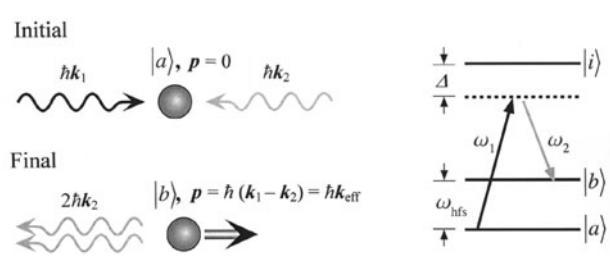

Figure 1. An atom undergoes a stimulated Raman transition. It absorbs a photon of frequency ω_1 and emits another one (via a stimulated emission) of frequency ω_2. As both photons carry momentum, and because momentum has to be conserved in the process, the atom receives a recoil momentum kick.

A. Peters,* K. Y. Chung and S. Chu: Physics Department, Stanford University, Stanford, CA 94305-4060, USA.
*Now with Fachbereich Physik, Universität Konstanz, M696, D-78457 Konstanz, Germany.

Table 1. Some gravimeters and their typical performance characteristics.

Type of gravimeter	Noise/ ($g\,\mathrm{Hz}^{-1/2}$)	Drift/ ($g\,\mathrm{day}^{-1}$)	Accuracy/g	Description and comments
Spring/mass system	1×10^{-10}	3×10^{-8}	N/A	A very stable, well-characterized and temperature-controlled spring supports a test mass against gravity. Variations in gravity change the spring's extension. Drift caused mostly by ageing of spring. Requires frequent recalibration. Very compact and transportable. [11]
Cryogenic gravimeter	$<10^{-12}$	2×10^{-10}	N/A	A superconducting sphere is levitated in a magnetic field and variations in gravity are counter-balanced using force feedback. Drift caused mostly by changing mass of sphere due to cryo-pumping. Vibration-induced flux-jumps in superconductor cause gravity offsets when instrument is moved. Lowest noise of all gravimeters, limited by environment. [11–13]
Falling corner-cube	5×10^{-8}	–	2×10^{-9}	A laser interferometer monitors the motion of a freely falling corner-cube retroreflector. Uses "super-spring" to eliminate effect of high-frequency vibrations. Noise depends strongly on site and drop rate. Quoted here is the noise obtained during a measurement at Stanford for a drop rate of 1/15 Hz. This drop rate is used instead of the maximum drop rate of 1 Hz to limit mechanical wear and prevent vibration-induced accuracy degradation. [11, 14, 15]
Atom interferometer	2×10^{-8}	–	3×10^{-9}	See text.

propagating beams) and an effective frequency $\omega_{\mathrm{eff}}=\omega_1-\omega_2$, where $\boldsymbol{k}_i$ and ω_i are the equivalent quantities for the individual laser beams.

(b) Absorption and stimulated emission of photons during a Raman pulse can change the momentum of the atom by $\hbar\boldsymbol{k}_{\mathrm{eff}}$ and, simultaneously, its internal state. The transition probability depends on the pulse area and can be adjusted to create either beam splitters or mirrors.

(c) The quantum mechanical phase of the resulting superposition state depends on the local Raman phase $\phi_i=k_{\mathrm{eff}}z_i-\omega_{\mathrm{eff}}t_i$.

Figure 2 shows how a sequence of three such Raman pulses is used to split, reflect and recombine an atom while simultaneously changing its internal state. At the end of this sequence the fraction of the atoms in one of the states is detected. The result is an oscillatory function of the interferometer path difference, which, among other factors, depends on the gravitational acceleration.

The total phase difference between paths A and B can be divided into two parts. The first contribution describes the periods of free evolution between laser pulses and is given by

$$\Delta\phi_{\mathrm{path}}=(S_{\mathrm{cl}}^{\mathrm{B}}-S_{\mathrm{cl}}^{\mathrm{A}})/\hbar \tag{1}$$

in the limit where the classical action

$$S_{\mathrm{cl}}=\int_0^{2T}\mathcal{L}[z(t),\,\dot{z}(t)]\,\mathrm{d}t \tag{2}$$

along each path is much greater than $\hbar$ [18]. For uniform gravitational fields ($\mathcal{L}=mv^2/2-mgz$) this contribution vanishes.

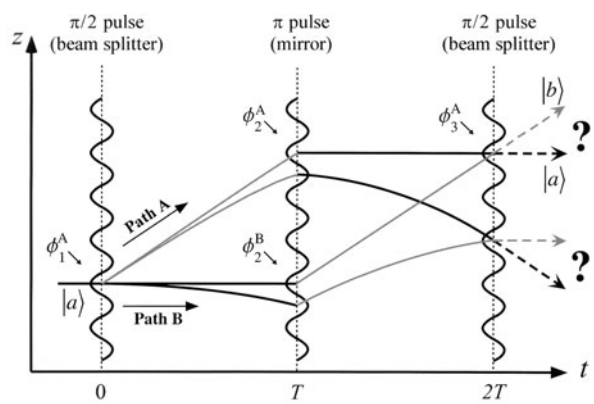

Figure 2. Basic Mach-Zehnder-type atom interferometer with (curved lines) and without (straight lines) gravity. The atom can either be in the internal state $|a\rangle$ (dark) or $|b\rangle$ (light). The lines represent the classical trajectories originating from one of the space-time points comprising the initial wave packet.

The second contribution is due to the interaction with the Raman beams. Whenever the state of the atom changes during such an interaction, it acquires an additional phase $\phi_i=k_{\mathrm{eff}}z_i-\omega_{\mathrm{eff}}t_i$, where z_i is the position of the atom at time t_i. The sign of the phase depends on the initial state of the atom. Tracing all the state changes we find a phase difference of

$$\Delta\phi_{\mathrm{light}}=(\phi_1^{\mathrm{A}}-\phi_2^{\mathrm{A}})-(\phi_2^{\mathrm{B}}-\phi_3^{\mathrm{A}}). \tag{3}$$

Without a gravitational field the trajectories are straight lines and the inherent symmetry of the situation (Figure 2) leads to $\Delta\phi=0$. The introduction of a gravitational field breaks the symmetry. The atom now falls three times as far during transit in the second half

of the interferometer as in the first half and we find a phase shift proportional to the gravitational acceleration:

$$\Delta\phi = k_{\mathrm{eff}} g T^2. \tag{4}$$

This simple equation describes the interferometer extremely well and has to be modified only slightly to account for gravity gradients (Section 2) and finite-length Raman pulses. The equation also indicates that it is highly desirable to make the time T between the interferometer pulses as long as possible, as the sensitivity of the measurements scales quadratically with T. One of the greatest improvements made since the first measurements in 1991 has been the implementation of an active vibration isolation system that allowed us to increase T to at least $160\,\mathrm{ms}$ without loss of interferometer phase contrast.

Our interferometer uses laser-cooled caesium atoms in an atomic fountain and Raman lasers that operate close to the D_2 line at $852\,\mathrm{nm}$. For a typical time $T = 160\,\mathrm{ms}$ between Raman pulses, the gravitational acceleration of an atom causes a phase shift of approximately $3.8 \times 10^6\,\mathrm{rad}$.

Details of our atom interferometer are given in the following sections. Section 2 presents a fuller theoretical analysis including the effects of gravitational gradients. Section 3 describes the experimental apparatus. Section 4 gives an overview of the experimental results, including long-term gravity measurements and a direct comparison with a falling corner-cube gravimeter. Section 5 contains an extensive analysis of potential systematic errors and how they affect the accuracy of the measurement. Finally, Section 6 summarizes the results and considers the prospects for future improvements and applications.

2. Theory

Gravity gradients have been neglected in earlier analyses of the atom interferometer [4, 19] but become important at a targeted measurement accuracy of $10^{-9}\,g$. This section presents a path integral treatment that includes a linear gravity gradient.

The path integral description of quantum mechanics [18], supplemented by a set of rules to describe the effects of infinitesimally short beam-splitter and mirror pulses, provides a natural framework for analysing light-pulse atom interferometers. The phase terms due to the atom/light interaction are summarized in Table 2 and illustrated in Figure 3.

Kasevich and Chu used this approach as one method to analyse their interferometer in 1991 [4]. A very similar approach was also developed by Bordé [20, 21] and in a tutorial paper by Storey and Cohen-Tannoudji [19]. We now extend these treatments to include the effect of a linear gravity gradient.

We assume a two-level system with wave functions Ψ_a and Ψ_b, and the initial state to be a plane wave with

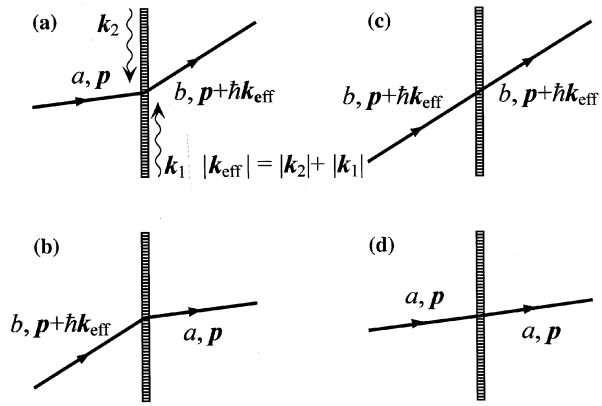

Figure 3. Basic processes during atom/light interaction. (a) When changing from internal state $|a\rangle$ to $|b\rangle$ the atom absorbs one photon from Raman beam 1 and emits another one co-propagating with Raman beam 2. Its momentum changes by $+\hbar \boldsymbol{k}_{\mathrm{eff}}$. (b) When changing from internal state $|b\rangle$ to $|a\rangle$ the atom absorbs one photon from Raman beam 2 and emits another one co-propagating with Raman beam 1. Its momentum changes by $-\hbar \boldsymbol{k}_{\mathrm{eff}}$. (c) and (d) The atom may also stay in its original state.

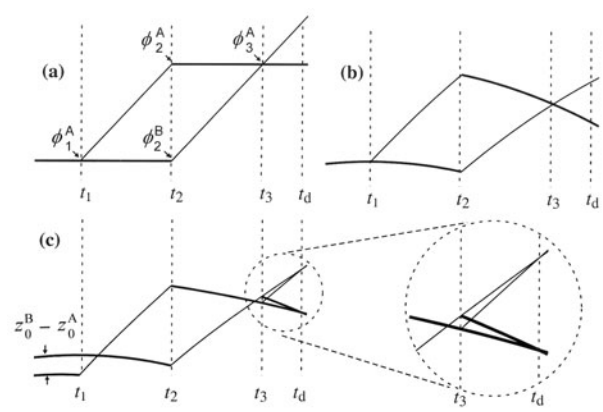

Figure 4. Classical interferometer paths. (a) No gravitational field, linear trajectories. (b) Uniform gravitational field, parabolic trajectories. After the final beam-splitter pulse at time t_3 both paths overlap exactly, for both possible outcomes. (c) Gravitational field with linear gravity gradient, hyperbolic trajectories. Interference at time t_{d} demands slightly different initial positions at time t_1 (highly exaggerated here).

momentum p_0 and a total energy E_0:

$$\Psi_a(z_0, t_0) = \frac{1}{\sqrt{2\pi\hbar}} \exp\left[\frac{i(p_0 z_0 - E_0 t_0)}{\hbar}\right],$$

$$\Psi_b(z_0, t_0) = 0. \tag{5}$$

We have to consider the two classical interferometer paths leading to the detection space-time point $(z_{\mathrm{d}}, t_{\mathrm{d}})$ as shown in Figure 4. The net phase difference can be calculated using the following algorithm:

- Find the two classical paths Γ_{A} and Γ_{B} for initial momentum p_0, taking into account the momentum

Table 2. Rules for atom/light interaction. (z_i, t_i) is the space-time point of the interaction. The quantities k, ω and ϕ denote the wave vector, frequency and phase of the light field. Note that the only change for stimulated Raman transitions – besides using the effective quantities $k_{\rm eff}$, $\omega_{\rm eff}$ and $\phi_{\rm eff}$ – is the appearance of additional phase factors $\phi^{\rm ac}_{\rm avg}$ and $\phi^{\rm ac}_{\rm diff}$ due to ac Stark shifts [16].

	Internal state	Momentum	Phase factor ϕ_i
1-photon transitions	$a \to b$	$\boldsymbol{p} \to \boldsymbol{p} + \hbar\boldsymbol{k}$	$+(k_i z - \omega t_i - \phi) + \pi/2$
	$b \to a$	$\boldsymbol{p} + \hbar\boldsymbol{k} \to \boldsymbol{p}$	$-(k_i z - \omega t_i - \phi) + \pi/2$
	$a \to a$	$\boldsymbol{p} \to \boldsymbol{p}$	0
	$b \to b$	$\boldsymbol{p} + \hbar\,\boldsymbol{k} \to \boldsymbol{p} + \hbar\,\boldsymbol{k}$	0
Raman transitions	$a \to b$	$\boldsymbol{p} \to \boldsymbol{p} + \hbar\,\boldsymbol{k}_{\rm eff}$	$\phi^{\rm ac}_{\rm avg} + (k_{\rm eff} z_i - \omega_{\rm eff} t_i - \phi_{\rm eff}) + \pi/2$
	$b \to a$	$\boldsymbol{p} + \hbar\,\boldsymbol{k}_{\rm eff} \to \boldsymbol{p}$	$\phi^{\rm ac}_{\rm avg} - (k_{\rm eff} z_i - \omega_{\rm eff} t_i - \phi_{\rm eff}) + \pi/2$
	$a \to a$	$\boldsymbol{p} \to \boldsymbol{p}$	$(\phi^{\rm ac}_{\rm avg} - \frac{1}{2}\phi^{\rm ac}_{\rm diff})$
	$b \to b$	$\boldsymbol{p} + \hbar\,\boldsymbol{k}_{\rm eff} \to \boldsymbol{p} + \hbar\boldsymbol{k}_{\rm eff}$	$(\phi^{\rm ac}_{\rm avg} + \frac{1}{2}\phi^{\rm ac}_{\rm diff})$

changes due to the interactions with the light field. The paths must meet at the time of detection $t_{\rm d}$ but can originate from different points $z_0^{\rm A}$ and $z_0^{\rm B}$.

- Calculate the classical action

$$S_{\rm cl} = \int_{t_0}^{t_f} \mathcal{L}\left[z(t), \dot{z}(t)\right] {\rm d}t \tag{6}$$

along each path and find the phase difference

$$\Delta\phi_{\rm path} = (S_{\rm cl}^{\rm B} - S_{\rm cl}^{\rm A})/\hbar. \tag{7}$$

- For each path add up the additional phase shifts due to the interactions with the light fields (see Table 2) and find the phase difference

$$\Delta\phi_{\rm light} = \sum_{\rm path\,B} \phi_j^{\rm B} - \sum_{\rm path\,A} \phi_i^{\rm A}. \tag{8}$$

- Account for the different initial points by defining the quantity

$$\Delta\phi_{\rm split} = p_0\,(z_0^{\rm B} - z_0^{\rm A})/\hbar. \tag{9}$$

- Calculate the total phase difference as

$$\Delta\phi_{\rm tot} = \Delta\phi_{\rm path} + \Delta\phi_{\rm light} + \Delta\phi_{\rm split}. \tag{10}$$

For a gravitational field with a linear gradient, the Lagrangian is given by

$$\mathcal{L} = m\dot{z}^2/2 - m g_0 z + m\gamma z^2/2, \tag{11}$$

where z denotes altitude, g_0 the gravitational acceleration at $z = 0$, and γ the linear gravitational gradient. The signs of g_0 and γ are chosen to be positive for normal conditions (objects accelerate downwards, magnitude of acceleration decreases with altitude).

Solving the Euler-Lagrange equation we find the exact classical path for initial position z_0 and velocity v_0:

$$\begin{aligned} z(t) =& \frac{g_0}{\gamma} + \left(z_0 - \frac{g_0}{\gamma}\right)\cosh\left(t\sqrt{\gamma}\right) + \\ & \frac{v_0}{\sqrt{\gamma}}\sinh\left(t\sqrt{\gamma}\right) \\ \dot{z}(t) =& \sqrt{\gamma}\left(z_0 - \frac{g_0}{\gamma}\right)\sinh\left(t\sqrt{\gamma}\right) + \\ & v_0\cosh\left(t\sqrt{\gamma}\right). \end{aligned} \tag{12}$$

Following the algorithm outlined in (6) to (10) we obtain an exact analytical solution using computer algebra. Here we present only the first few terms of its power series expansion in the gradient γ:

$$\begin{aligned} \Delta\phi_{\rm tot} =& \left(m v_{\rm rec} T^2/\hbar\right) g_0 + \gamma\left(m v_{\rm rec} T^2/\hbar\right) \times \\ & \left(\frac{7}{12} g_0 T^2 - \bar{v}_0 T - z_0\right) + \\ & \gamma^2 T^2 \left(m v_{\rm rec} T^2/\hbar\right) \times \\ & \left(\frac{31}{360} g_0 T^2 - \frac{1}{4}\bar{v}_0 T - \frac{7}{12} z_0 - \frac{1}{2} v_{\rm rec} T - \right. \\ & \left. \frac{1}{2} v_{\rm rec} T_{\rm det}\right) + \cdots . \end{aligned} \tag{13}$$

We may rewrite this as

$$\begin{aligned} g_{\rm m} =& \Delta\phi/(m v_{\rm rec} T^2/\hbar) \\ =& g_0 + \gamma\left\{\frac{7}{12} g_0 T^2 - \bar{v}_0 T - z_0\right\} + \\ & \gamma^2 T^2 \left\{\frac{31}{360} g_0 T^2 - \frac{1}{4}\bar{v}_0 T - \frac{7}{12} z_0 - \right. \\ & \left. \frac{1}{2} v_{\rm rec} T - \frac{1}{2} v_{\rm rec} T_{\rm det}\right\} + \cdots . \end{aligned} \tag{14}$$

In these equations T is the time interval separating the interferometer pulses, $T_{\rm det}$ is the time delay between the final pulse and the actual detection of the atomic state and t_0 denotes the time of the first pulse. The quantities z_0 and v_0 are the initial position and velocity for path B. The recoil velocity $v_{\rm rec} = \hbar k_{\rm eff}/m$ is directly proportional to the effective wave vector of the Raman beams and also determines the maximum separation of the two interferometer paths. To simplify the equations slightly we also introduce the average velocity $\bar{v}_0 = v_0 + v_{\rm rec}/2$ of the two different paths just after the first Raman pulse.

An interesting feature of this result is that the phase shift depends on the detection delay $T_{\rm det} = t_{\rm d} - t_3$. This comes about because the separation of the initial points of the two paths, $z_0^{\rm A}$ and $z_0^{\rm B}$ (see algorithm and Figure 4), is a function of $T_{\rm det}$. However, for typical experimental parameters (see below) the separation is only 1.7 Å and the effect is extremely small ($< 10^{-16}\,g$).

We can now evaluate the importance of the various terms. For the typical experimental parameters

$$\begin{aligned} &g_0 \simeq 9.8\,{\rm m\,s^{-2}}, \quad \gamma \simeq 2.9\times 10^{-6}\,{\rm s^{-2}},\\ &z_0 \simeq 0\,{\rm m}, \quad \bar{v}_0 \simeq 1.57\,{\rm m\,s^{-1}},\\ &v_{\rm rec} \simeq 7.0\times 10^{-3}\,{\rm m\,s^{-1}},\\ &T \simeq 0.16\,{\rm s}, \quad T_{\rm det} \simeq 0.16\,{\rm s}, \end{aligned}$$

we find

$$g^{(1)} = 3.1\times 10^{-8}\,g_0$$

and

$$g^{(2)} = 9.2\times 10^{-16}\,g_0,$$

where $g^{(n)}$ denotes the n-th order term in γ. At our targeted accuracy of $10^{-9}\,g$ we can therefore safely neglect terms of second order and higher.

We have also calculated the interferometer phase shift using a perturbative treatment [22, 19]. If we treat only the gradient term as a perturbation and integrate the perturbation Lagrangian $\mathcal{L}_1 = m\gamma z^2/2$ over the unperturbed parabolic paths, then we find

$$g_{\rm m} = g_0 + \gamma\left\{\frac{7}{12}\,g_0T^2 - \bar{v}_0T - z_0\right\}, \tag{15}$$

which replicates the exact path integral result up to first order in γ.

We cannot, however, simultaneously treat gravity and gradient as perturbations. In this case integrating the perturbation Lagrangian $\mathcal{L}_1 = m\gamma z^2/2 - mg_0z$ over the unperturbed straight paths leads to

$$g_{\rm m} = g_0 - \gamma\,\{\bar{v}_0T + z_0\}, \tag{16}$$

which differs significantly from the previous result (about $5\times 10^{-8}\,g$ for typical experimental parameters).

Since the gravity-induced interferometer phase shift can be expressed without the quantum scale factor $\hbar/m$ as in (4) and (13), it is also possible to ignore the quantum nature of the atom. One can model it as a classical point particle that carries an internal clock, measures the local phases $\phi_i = k_{\rm eff}z(t_i) - \omega_{\rm eff}t_i - \phi_0$ of the light field during the interferometer pulses and finally calculates $g_{\rm m} = (\phi_3 - 2\phi_2 + \phi_1)/k_{\rm eff}T^2$. The power series expansion

$$\begin{aligned} g_{\rm m} = g_0 + \gamma\left\{\frac{7}{12}\,g_0T^2 - v_0T - z_0\right\} + \\ \gamma^2T^2\left\{\frac{31}{360}\,g_0T^2 - \frac{1}{4}\,v_0T - \frac{7}{12}\,z_0\right\} + \ldots \end{aligned} \tag{17}$$

is identical to the exact path integral result up to first order and even the second-order term only differs by $\frac{1}{2}\gamma^2T^2v_{\rm rec}(T+T_{\rm det})$, which is exceedingly small (about $3\times 10^{-16}\,g$) for standard experimental parameters and vanishes completely in the limit of infinite atomic mass ($v_{\rm rec} = 0$).

3. Experimental apparatus

The overall experimental apparatus (Figure 5) can be divided into three subsystems:

(a) The atomic fountain apparatus, which provides a well-defined source of laser-cooled atoms for the interferometer. It also allows the detection of the final state of the atoms at the output of the interferometer.

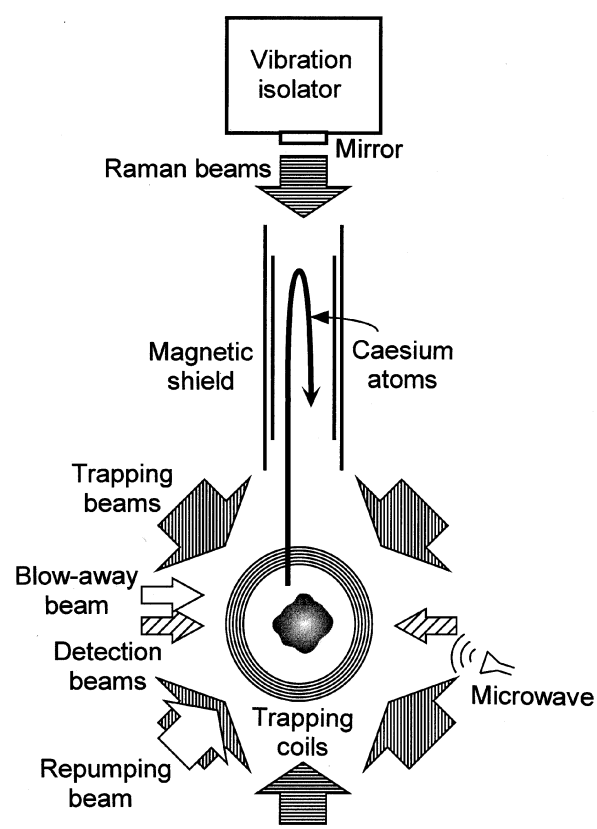

Figure 5. Overview of experimental set-up.

(b) The system for implementing the actual light pulse interferometer, including the phase-locked diode lasers and the equipment for controlling the frequency and timing of the Raman pulses. Other components are the Raman beam optics and the magnetic shielding of the measurement region.

(c) The vibration isolation system, which allows the large interferometer pulse separations essential for obtaining our current measurement accuracy and precision. Another part of this system controls the tilt of the optical table to guarantee the proper vertical alignment of the measurement axis.

A more detailed discussion can be found in [23].

3.1 Atomic fountain

3.1.1 Magneto-optic trap (MOT)

The caesium atoms are initially extracted from a low-pressure background vapour and loaded into a magneto-optic trap (MOT) [24]. After $600\,\mathrm{ms}$ of loading the trap contains 5×10^8 atoms in a cloud approximately $5\,\mathrm{mm}$ in diameter.

Our MOT closely resembles the one described in [24, 25], except that it does not use a pure vapour cell design. The relatively high pumping speed of our $30\,\mathrm{l/s}$ ion pump, in combination with the placement of caesium reservoir and pump on different sides of the trapping region, leads to a partial pressure of caesium which is neither uniform nor in equilibrium with the cooled reservoir ($+15\,^\circ\mathrm{C}$ would correspond to $\sim 7 \times 10^{-5}\,\mathrm{Pa}$ ($5 \times 10^{-7}\,\mathrm{Torr}$) [26]). Instead, we find (using absorption measurements) a partial pressure of caesium of less than $1 \times 10^{-8}\,\mathrm{Pa}$ in most parts of the vacuum chamber, substantially lower than the $\sim 5 \times 10^{-8}\,\mathrm{Pa}$ pressure due to other gases. We therefore assume that the atoms, which enter through a 3.8 cm diameter tube, effectively form a large-diameter atomic beam through the trapping region.

We use three pairs of counter-propagating trapping beams in a 1,1,1 configuration (threefold symmetry around the vertical axis, three beams directed upwards, three downwards). The magnetic trapping field is generated by water-cooled coils wound on the outside of the vacuum chamber and with an axis parallel to one of the beam pairs. The standard operating field gradient is $6\,\mathrm{G/cm}$. Each trapping beam is circularly polarized, $4\,\mathrm{cm}$ in diameter ($1/e^2$ intensity contour), has a typical intensity of $15\ \mathrm{mW/cm^2}$ and a frequency detuned $20\,\mathrm{MHz}$ to the red of the $6S_{1/2}, F = 4 \leftrightarrow 6P_{3/2}, F' = 5$ transition at $852\,\mathrm{nm}$. One of the trapping beams is nearly overlapped with a repumping beam having an intensity of about $1.5\,\mathrm{mW/cm^2}$ and is on resonance with the $6S_{1/2}, F = 3 \leftrightarrow 6P_{3/2}, F' = 4$ transition.

3.1.2 The laser system

All laser beams (trapping, detection, 4-5 blow-away) close to $6S_{1/2}, F = 4 \leftrightarrow 6P_{3/2}$ excited-state transitions (see Figure 6 for energy-level diagram) of the caesium D_2 line at $852\,\mathrm{nm}$ are derived from an argon-ion laser-pumped Ti:sapphire ring laser (Coherent 899-21) with a typical output power of 2.0 W to 2.5 W (Figure 7). This laser is stabilized relative to the $F = 4 \leftrightarrow F' = 4$, $F = 4 \leftrightarrow F' = 5$ crossover resonance using a standard FM saturation spectroscopy lock [27]. Because of concerns about vibrations, the Ti:sapphire and argon-ion lasers are installed on a separate table and the light is routed to the main optical table using a 2 m long single-mode, polarization-preserving optical fibre with pre-installed collimation lenses (Oz optics). Using a telescope for proper mode matching, we can achieve a

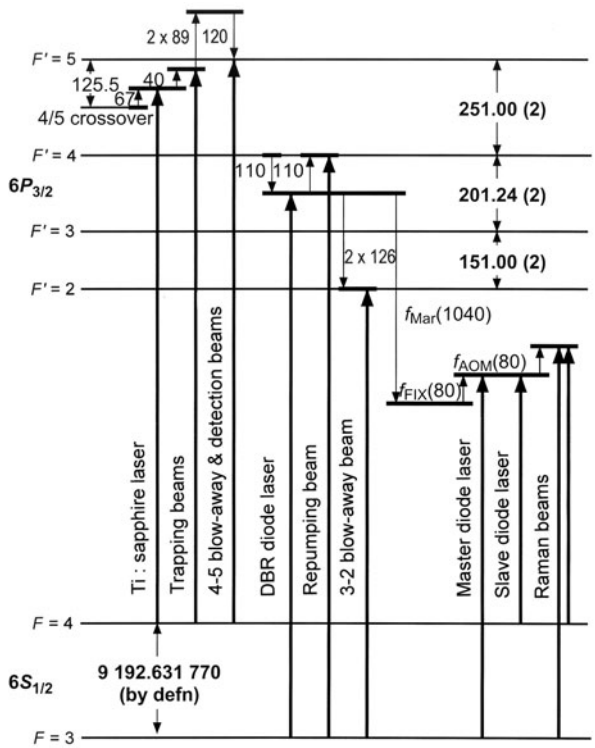

Figure 6. Atomic energy levels and laser frequencies. All laser frequencies are referenced either to the $F = 4 \leftrightarrow F' = 4/F' = 5$ "crossover" resonance or the $F = 3 \leftrightarrow F' = 4$ resonance. Numerical values are in megahertz.

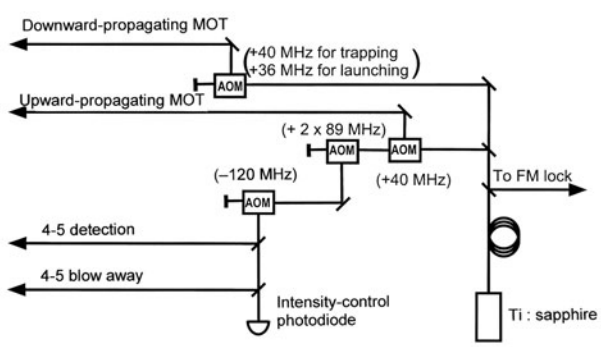

Figure 7. Beams derived from the Ti:sapphire laser.

very high coupling efficiency and deliver 75 % of the laser output power to the optical table.

All laser beams (repumping, 3-2 blow-away, Raman frequency reference) close to $6S_{1/2}$, $F = 3 \leftrightarrow 6P_{3/2}$ excited-state transitions are derived from a distributed-Bragg-reflector (DBR) diode laser (Figure 8). This laser (SDL-5712-H1, 100 mW, single transverse and longitudinal mode, ~5 MHz linewidth) is stabilized relative to the $F = 3 \leftrightarrow F' = 4$ resonance using saturation spectroscopy and a fast dither lock.

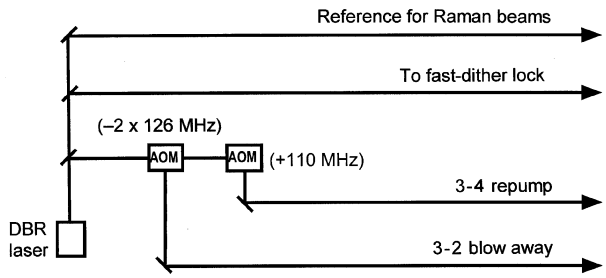

Figure 8. Beams derived from the distributed-Bragg-reflector (DBR) laser.

3.1.3 Launching the atoms

After turning off the magnetic trapping fields, 7 ms are allowed for the decay of eddy currents (Figure 9). The atoms are launched vertically using moving optical molasses [28]. The frequency of the upper three trapping beams (travelling downwards) is shifted down by ~4.0 MHz. This is equivalent to shifting the frequency of the upper and lower beams in opposite directions by 2.0 MHz each (the 2.0 MHz change in the average detuning is negligible compared with the original 20 MHz). For the next 2 ms the atoms are allowed to accelerate and equilibrate at a new velocity of ~300 cm/s in the moving molasses. This velocity allows them to reach a maximum height of 46 cm above the trap. The temperature of the atoms after the launch is ~1.5 μK or less, well below the usual steady-state minimum temperature of 2.5 μK [29]. This is achieved by using far-detuned (60 MHz), low-intensity (1/3 of the MOT intensity) optical molasses during the last stage of the launch and by ramping down the laser intensity over a period of 0.4 ms, allowing the atoms to adiabatically cool [30]. The repumping light is shut off last to guarantee that all the atoms are in the $F = 4$ hyperfine state. The lowest temperatures are usually achieved by slightly misaligning the trapping beams to obtain a pancake-shaped MOT, compressed along the symmetry axis of trapping coils.

3.1.4 Atomic state preparation

After pulsing on a vertical magnetic bias field of ~500 mG to define a quantization axis, a ~100 μs long microwave π pulse (emitted from a horn antenna) transfers all the atoms in the $F = 4$, $m_F = 0$ state to the $F = 3$, $m_F = 0$ state with very high efficiency (typically 98 %). The remaining atoms are then blown away by a 1 ms long pulse of light in resonance with the $F = 4 \leftrightarrow F' = 5$ closed optical transition (Figure 10). A 380 μs long, velocity-selective Raman π pulse transfers atoms in a narrow slice of the velocity distribution (FWHM ≈ 0.1 cm/s) back into the $F = 4$, $m_F = 0$ state. The remaining atoms are then blown away using light in resonance with the $F = 3 \leftrightarrow F' = 2$ closed transition. Another microwave π pulse, followed by a final 4-5 blow-away pulse for clean-up, transfers the atoms back to the $F = 3$, $m_F = 0$ state. In principle this elaborate sequence could be replaced by a single velocity-selective Raman π pulse, but at our Raman detuning of $\Delta = 0.99$ GHz, residual spontaneous processes would then cause a large fraction (~60 %) of background atoms.

This selection sequence results in a very well-defined sample of $\sim 5 \times 10^6$ caesium atoms in the $6S_{1/2}$, $F = 3$, $m_F = 0$ state with a substantially reduced vertical velocity spread, corresponding to a 1-D temperature of only ~10 nK. The sample then enters a magnetically shielded region which encloses the top

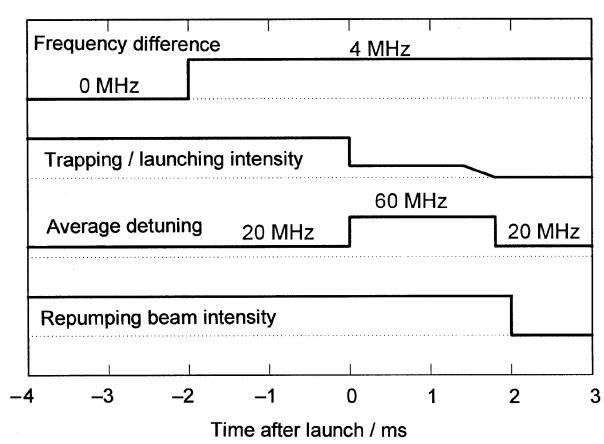

Figure 9. Launch sequence.

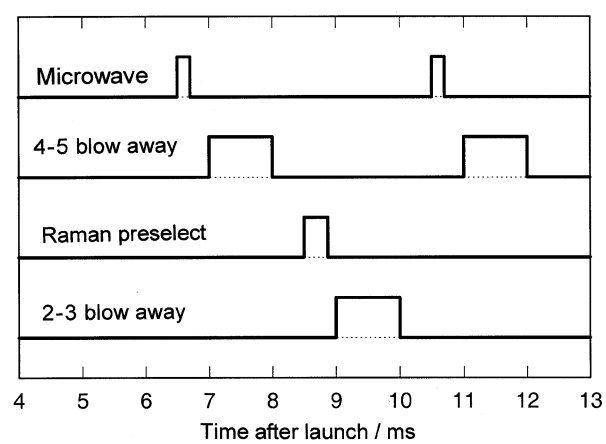

Figure 10. Atomic-state preparation sequence.

part of its trajectory and where the actual interferometer measurement takes place.

3.1.5 State-selective detection

The interferometer phase shift manifests itself as a modulation of the fraction of atoms in the $F = 4, m_F = 0$ state. This fraction is determined, after the atoms have left the magnetic shield and returned to the location of the original trap, using normalized fluorescence detection (Figure 11). The atom is illuminated by two identical probe pulses interspersed with an optical pumping pulse. The $500\,\mu\mathrm{s}$ long probe pulses are in resonance with the $6S_{1/2}, F = 4 \leftrightarrow 6P_{3/2}, F' = 5$ closed transition, circular polarized and have an actively stabilized intensity of typically $0.9\,\mathrm{mW/cm^2} \approx 0.8\,I_{\mathrm{sat}}$. The $100\,\mu\mathrm{s}$ long optical pumping pulse is in resonance with the $6S_{1/2}, F = 3 \leftrightarrow 6P_{3/2}, F' = 4$ transition and has an intensity of $\sim 1.5\,\mathrm{mW/cm^2}$. The fluorescence is imaged on to a photomultiplier tube (PMT) which detects 1 out of every 80 scattered photons ($\sim$5 photons per atom). The fluorescence during the first probe pulse is proportional to the number of atoms in the $F = 4$ state. The pumping pulse additionally transfers all atoms from the $F = 3$ to the $F = 4$ state. The fluorescence during the second probe pulse is therefore proportional to the number of all atoms.

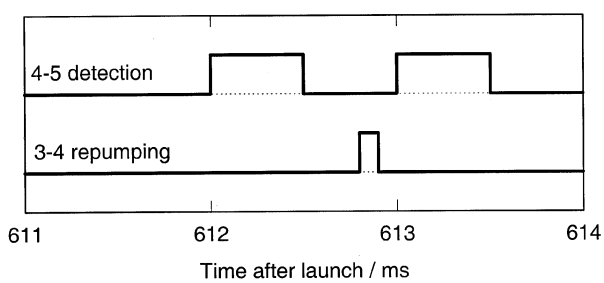

Figure 11. Detection sequence.

The horizontal probe beam is retroreflected to minimize the deflection of the atomic trajectories by light-scattering forces. Apertures in the probe beam and in front of the PMT can be used to detect only atoms in a restricted volume (Section 5.5.1).

Finally, after a short delay, the MOT is turned on again – starting the next measurement cycle $1.3\,\mathrm{s}$ after the previous one.

3.2 Interferometer

This section presents the system for generating the laser pulses forming the beam splitters and mirrors of the atom interferometer. The magnetic shielding assembly surrounding the measurement region is also described.

3.2.1 Phase-locked diode lasers

We generate the Raman light by phase-locking two external-cavity diode lasers [16, 31]. Their difference frequency and phase are then referenced to a stable rf source and can be controlled precisely (Figure 12).

The external-cavity lasers use laser diodes (SDL-5411-G1, GaAlAs, $100\,\mathrm{mW}$, single transverse and longitudinal mode) in Littrow configuration. They have short-term linewidths substantially smaller than $1\,\mathrm{MHz}$ and provide up to $80\,\mathrm{mW}$ of output power each. The slave laser features a fast, ac-coupled modulation input that bypasses the current controller to minimize propagation delays.

The beat note of the two overlapped laser beams ($f_{\mathrm{Raman}} \approx 9.192\,\mathrm{GHz}$) is detected by a fast photodiode (Antel S2), amplified and mixed down with a local oscillator at frequency $f_{\mathrm{DRO}}(\approx 9.140\,\mathrm{GHz})$ to a convenient intermediate frequency $f_{\mathrm{SRS}}(\approx 52\,\mathrm{MHz})$, which is then amplified once again and used as the input for three different feedback paths. The fast path feeds the output signal of an analogue phase detector directly back into the fast modulation input of the slave laser. A compensation filter is used to maximize the feedback bandwidth. The intermediate and slow paths both use a digital phase-detector circuit that has better low-frequency properties than the analogue mixer and provides a bigger capture range for the phase-locked loop. The intermediate path uses the dc-modulation input of the diode laser current controller. The slow path uses a piezo to adjust the length of the external diode-laser cavity. The crossover points between the three feedback bands are typically $25\,\mathrm{Hz}$ and $550\,\mathrm{kHz}$. The total feedback bandwidth is approximately 2.0 MHz to 2.5 MHz. With the phase-lock on, we typically find $99.8\,\%$ of the total laser power in the carrier of the beat-note signal.

3.2.2 Frequency control

The relative phase of the Raman beams provides the reference against which all interferometer phase shifts are measured. Any noise in the Raman phase will directly translate into noise of the gravity data and any failure to control it properly could result in systematic errors in the measured gravity value.

For most of our interferometer measurements we compensate for the acceleration-induced Doppler shift ($\alpha \approx 23.0\,\mathrm{MHz/s}$) by switching between three fixed Raman frequencies that differ by an amount Δf (see Figure 13). For this method to work, it is essential that the frequency changes are phase continuous and that the time between them equals the interferometer pulse spacing, T. The exact delay τ_{d} between pulses and frequency changes, on the other hand, is not important. The total interferometer phase shift is then given by

$$\Delta\phi_{\mathrm{tot}} = \Delta\phi_g - \Delta\phi_{\mathrm{rf}}, \tag{18}$$

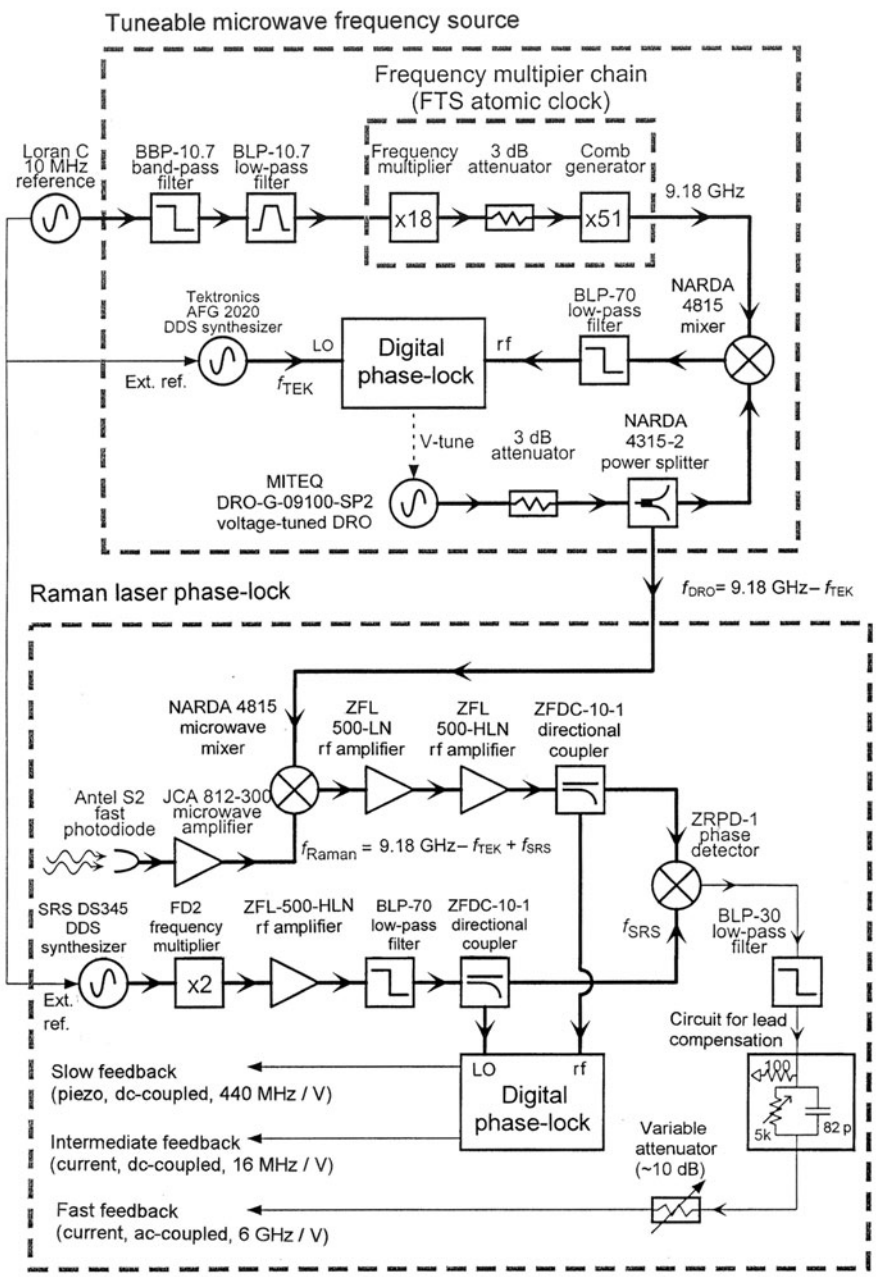

Figure 12. Raman laser phase-lock and system for controlling the Raman difference frequency and phase.

where

$$\Delta\phi_g = k_{\rm eff} g T^2 \quad \text{and} \quad \Delta\phi_{\rm rf} = 2\pi\Delta f T \tag{19}$$

are the phase shift due to gravity and the modification due to changes in the Raman frequency, respectively. If $\Delta\phi_{\rm tot}$ is made to vanish by adjusting Δf, then the value of g is given by the relation

$$g = \frac{2\pi\Delta f}{k_{\rm eff} T}. \tag{20}$$

For interferometer finite-length pulses the proper definition of the pulse spacing is $T = T_0 + 2\tau$, where τ is the length of a $\pi/2$ pulse and T_0 the time of free evolution between pulses [23].

The system for controlling the Raman frequency difference and phase relationship is shown in Figure 12. All frequencies are derived from a 10 MHz Loran-C (SRS FS 700-01) frequency standard, which itself is ultimately referenced to timing signals from the US Naval Observatory's atomic clock. The 10 MHz reference frequency is multiplied up to 9.180 GHz using a two-stage frequency-multiplier chain, identical to the ones used in Frequency and Time Systems (FTS) atomic clocks. The first stage provides a clean, single-frequency output signal at 180 MHz. The second stage uses a comb

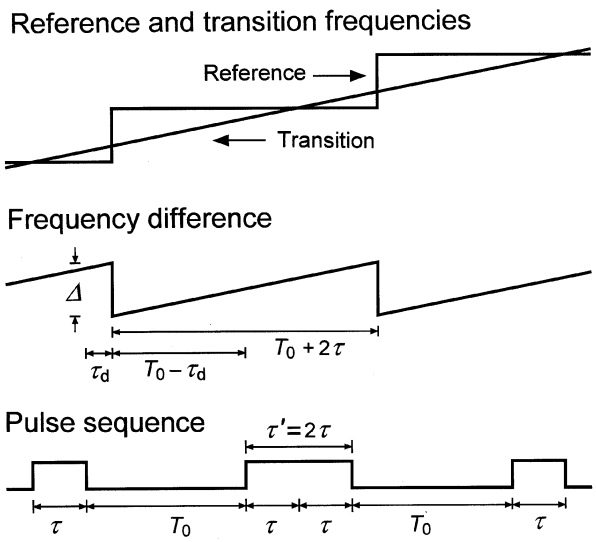

Figure 13. Timing relationship between interferometer pulses and frequency changes. The acceleration-induced Doppler shift causes a continuous change of the atomic transition frequency. To compensate, the Raman frequency difference is adjusted in discrete steps. The exact delay τ_d of these frequency changes relative to the Raman pulses is not important, but they have to be spaced by exactly $(T_0 + 2\tau)$.

generator to produce many high-order harmonics of the 180 MHz input frequency. One particular component, at 9.180 GHz, serves as our microwave frequency reference.

We generate a tuneable microwave source by phase-locking a dielectric resonator oscillator (DRO) to the 9.180 GHz reference signal with an adjustable, synthesized offset frequency f_TEK of typically 40 MHz. This signal is used to mix down the photodiode beat-note signal to generate an intermediate frequency. The Raman laser phase-lock then keeps this frequency equal to another synthesized reference frequency f_SRS of typically 52.631 770 MHz. Overall, the Raman laser frequency difference is given by the formula

$$f_\mathrm{Raman} = 9.18\,\mathrm{GHz} - f_\mathrm{TEK} + f_\mathrm{SRS}. \tag{21}$$

We generate the frequency steps by changing f_TEK, since the DRO phase-lock has only one feedback path and behaves in a predictable way. The diode laser phase-lock, on the other hand, with its multiple feedback paths, is very complex and introduces phase shifts that are difficult to predict.

The frequency f_TEK itself is generated by a Tektronics AFG-2020 DDS synthesizer (250 MHz sampling rate) that is phase-locked to the Loran-C frequency reference. To implement the required frequency steps, we use the synthesizer's frequency shift keying (FSK) modulation mode. The frequency changes are synchronized with the phase of the output waveform and can be set with a resolution of 10 µs. It is essential that the time between frequency changes equals the interferometer pulse spacing T.

It is possible to perform the experiment by switching between only three frequencies. However, the 30-bit frequency resolution of the synthesizer ($\sim$0.23 Hz) would then only allow a relatively coarse step size of $\sim$0.23 rad when scanning interferometer fringes. We therefore often use a modified scheme that achieves a smaller step size by switching between more than three frequencies [23].

We also have the option of compensating for the acceleration-induced Doppler shift during each individual interferometer pulse. This is necessary for the 380 µs long velocity preselection pulse (Section 3.1.4), where the acceleration during the time of the pulse becomes significant ($\alpha\tau^2 > 1$). This capability is also useful to test for possible systematic errors related to frequency chirps (Section 5.3.4).

We implement this compensation via the second control frequency, f_SRS. Instead of holding it fixed at 40 MHz, we use a frequency-doubled direct digital synthesis (DDS) synthesizer (SRS DS-345) to sweep it for 1 ms at the correct chirp rate. A second SRS synthesizer is used to trigger this process repeatedly every 2 ms. Both synthesizers are locked to the external 10 MHz Loran-C reference. As these chirps only span a short frequency interval and are identical for all pulses, the phase shifts in the diode-laser phase-lock circuitry are not a problem.

The absolute frequency of the Raman lasers has to be known to at least the targeted accuracy of the gravity measurement. The master laser frequency is therefore referenced, via the DBR repumping laser, to the $6S_{1/2}, F = 3 \leftrightarrow 6P_{3/2}, F' = 4$ transition of caesium. The beat note of the two lasers is detected by a fast photodiode and mixed with an adjustable frequency f_ADJ in the low gigahertz range. The resulting intermediate frequency is kept at f_FIX using a frequency lock (here simply a low-gain version of our digital phase-lock circuit). We typically use $f_\mathrm{ADJ} = 1.040$ GHz and $f_\mathrm{FIX} = 80$ MHz. The absolute frequency f_m of the master laser, including a final frequency shift $f_\mathrm{AOM} = 80$ MHz due to the switching acousto-optic modulator (AOM), is then given by

$$\begin{aligned} f_\mathrm{m} &= f_{3\leftrightarrow 4} - 110\,\mathrm{MHz} - f_\mathrm{ADJ} + f_\mathrm{FIX} + f_\mathrm{AOM} \\ &= f_{3\leftrightarrow 4} - 110\,\mathrm{MHz} - 1040\,\mathrm{MHz} + \\ &\quad 80\,\mathrm{MHz} + 80\,\mathrm{MHz}, \end{aligned} \tag{22}$$

where $f_{3\leftrightarrow 4}$ is the absolute frequency of the $6S_{1/2}$, $F = 3 \leftrightarrow 6P_{3/2}, F' = 4$ optical transition (see also Figure 6).

3.2.3 Optics and beam path

Figure 14 shows the optical system for Raman beam generation and delivery to the atoms. The elliptical beam profiles of the diode lasers are made circular, overlapped using a polarizing beam-splitting cube and then coupled into a 3.9 m long, polarization-preserving

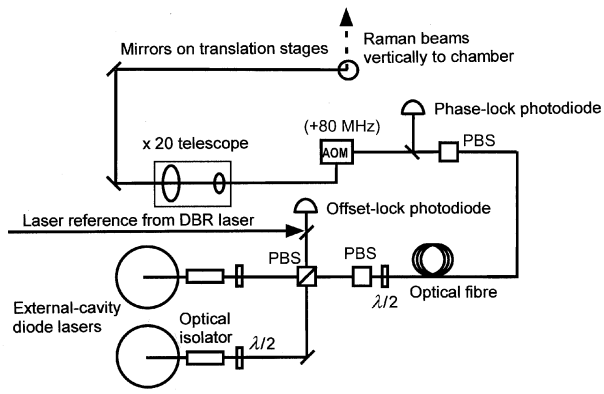

Figure 14. Raman laser system. PBS: polarizing beam splitter.

optical fibre (Oz optics). The output beam is first passed through another polarizing beam-splitting cube to ensure well-defined polarization. Then a small fraction is picked up by a beam splitter and sent to a fast photodiode to detect the beat note for the phase-lock. The collimated main beam then passes through an $80\,\mathrm{MHz}$ single-pass AOM used to generate Raman pulses of controlled intensity. The timing of the pulses is controlled by a set of synchronized pulse generators (SRS DG 535), one of which also triggers all the hardware involved in generating the Raman frequencies.

The first-order output beam of the AOM is then expanded to a diameter ($1/e^2$ intensity contour) of $\sim 2.7\,\mathrm{cm}$ and routed to the vacuum chamber using high-quality optics. Wavefront curvature and distortion are measured using a shear-plate collimation tester (Melles Griot 009-SPM-003). After passing through the vacuum chamber the Raman beams are retroreflected by a high-quality, vibration-isolated mirror (Section 3.3.1). They also pass twice through a high-quality $\lambda/4$-wave plate that rotates the polarization of the beams by $90\,^\circ$.

This set-up allows us to drive Doppler-sensitive Raman transitions in (*lin* $\perp$ *lin*) configuration, using one of the two frequency components in the upward-propagating beam together with the other, orthogonally polarized frequency component in the downward-propagating beam. All other combinations do not contribute because of polarization selection rules and detuning from the Raman resonance condition. It is also possible to drive Doppler-insensitive ($\sigma^+ - \sigma^+$) Raman transitions by inserting another $\lambda/4$-wave plate into the beam before it enters the vacuum chamber and blocking the retroreflection.

3.2.4 Interaction region

The whole interferometer sequence takes place while the atom is inside a magnetically shielded interaction region. The magnetic shield consists of four cylinders made from $1\,\mathrm{mm}$ thick Hipernom. All but the innermost cylinder are also fitted with end caps with 3.8 cm diameter holes. This shielding assembly should have theoretical radial and axial shielding factors of 1.6×10^9 and 2.9×10^7, respectively [32]. To generate a uniform, vertical bias field a 29 cm long solenoid is installed inside the innermost shield. It consists of two layers of $1\,\mathrm{mm}$ diameter, Kapton-coated wire which is wrapped on an aluminium core 3.8 cm in diameter. The solenoid can also be used for degaussing the two innermost layers of the magnetic shield. See Figure 29 for a measurement of the magnetic field homogeneity inside the interaction region.

3.3 Vibration isolation and tilt control

Any change in the relative paths of the two Raman beams due to vibrations will be seen as a spurious interferometer phase shift and noise in the g measurement. To prevent the interferometer fringes from washing out, vibrations have to be reduced to a level where the resulting phase shifts are much smaller than 1 rad. This translates into the requirement of allowing only displacements much smaller than a single optical wavelength during the interferometer drift time.

For interferometers using thermal atomic beams, the drift times are sufficiently short so that normal laboratory vibration-isolation tables with their resonance frequencies of typically $2\,\mathrm{Hz}$ are adequate. However, the long drift times (up to $2T = 320\,\mathrm{ms}$) used in our atomic fountain interferometer require the suppression of vibrations at frequencies well below $1\,\mathrm{Hz}$. There are also particularly strong geological background vibrations ("microseisms") at periods of approximately 5 s and 8 s. The vibration-isolation system for our experiment should therefore have a resonance frequency substantially below $0.1\,\mathrm{Hz}$.

3.3.1 Vertical active vibration isolation

In our set-up any vibrational noise added to the co-propagating Raman beams before they reach the measurement region is common-mode noise and cancels since only the frequency difference $\omega_1 - \omega_2$ is relevant for the interferometer. The retroreflection mirror is therefore the only element that needs to be isolated against vibrations. As the interferometer is mainly sensitive to vertical accelerations, a single-axis low-frequency vibration-isolation system is sufficient to achieve major reductions in measurement noise.

Passive vertical vibration isolation using a spring suspension is not possible at low frequencies. For a linear spring supporting a mass against a gravitational acceleration g, the resonance frequency is $f_0 = \sqrt{g/L}/(2\pi)$, where L is the spring extension from equilibrium. Achieving a resonance frequency of $0.1\,\mathrm{Hz}$ would thus require a prohibitively long spring extension of more than $25\,\mathrm{m}$. We have therefore constructed an active system that combines relatively short mechanical springs with an electronic feedback system to create an almost critically damped spring-mass system with an effective resonance frequency of $\sim 0.005\,\mathrm{Hz}$.

We now describe the basic theory, implementation and performance of the active vibration-isolation system. A more detailed discussion of a very similar active vibration isolation can be found in [33].

Assume that a mass m is suspended from a point rigidly connected to the ground using a linear spring with spring constant k and damping factor β (Figure 15). We want to isolate its motion $z(t)$ from any motion $z_{\mathrm{g}}(t)$ of the ground, both measured relative to a stable reference point. The system has the equation of motion

$$\ddot{z} + 2\,\zeta_0\omega_0\,(\dot{z} - \dot{z}_{\mathrm{g}}) + \omega_0^2\,(z - z_{\mathrm{g}}) = 0, \qquad (23)$$

where $\omega_0^2 = k/m$ is the natural resonance frequency and $\zeta_0 = \beta/(2m\omega_0)$ is the damping constant ($\zeta_0 = 1$ for a critically damped system). Our system typically has a natural resonance frequency of 1 Hz and very little damping ($\zeta_0 = 0.03$).

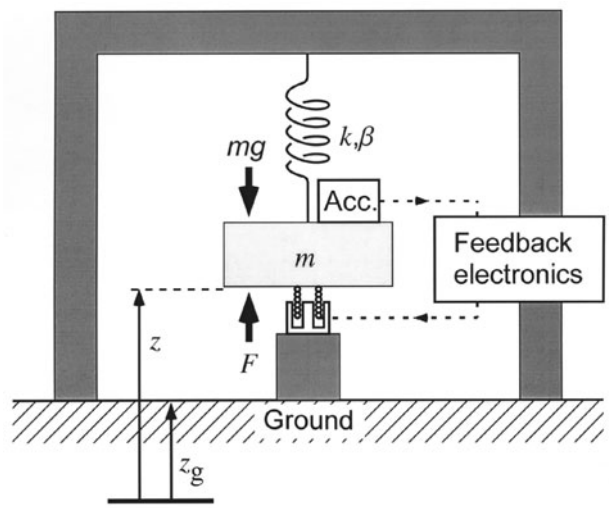

Figure 15. Schematic of the vibration isolator.

In order to lower the resonance frequency of the system we use a magnetic actuator to apply two additional forces, F_{a} and F_{v}, to the suspended mass. $F_{\mathrm{a}} = -mG\ddot{z}$ is proportional to the acceleration of the mass and $F_{\mathrm{v}} = -2m\omega_0 H\dot{z}$ is proportional to its velocity. Here G and H are constants determined by the feedback electronics and $\ddot{z}$ and $\dot{z}$ are both derived from the signal of an accelerometer mounted to the suspended mass. The resulting equation of motion

$$(1+G)\,\ddot{z} + 2\,\zeta_0\omega_0\,[(1+H)\,\dot{z} - \dot{z}_{\mathrm{g}}] + \omega_0^2\,(z - z_{\mathrm{g}}) = 0 \qquad (24)$$

shows that the system now has a new resonance frequency $\omega_1 = \omega_0/\sqrt{G+1}$ and a new damping factor $\zeta_1 = (\zeta_0 + H)/\sqrt{G+1}$. Increasing the gain G lowers both the resonance frequency and the damping. To compensate for the loss in damping, it is also necessary to increase the gain H.

In the real system the acceleration of the mass is measured by a low-noise, low-frequency force-feedback accelerometer (Guralp CMG-3V) and then digitized using a 16-bit data-acquisition board. The signal is then processed by a digital loop filter that implements the gains G and H (typically $G \approx 10\,000$, $H \approx 300$) but also contains additional compensators to maintain closed-loop stability at high ($\sim$80 Hz) as well as low ($\sim$0.001 Hz) frequencies [33]. To close the feedback loop the output signal is converted back into analogue form and used to control the current through the solenoid actuator.

The physical implementation of the vibration isolator (Figure 16) differs from that in [33] mainly in that it uses only two springs and has a different permutation of components. Placing the air piston on top makes the system more stable. The axis of the air piston must be parallel to $\boldsymbol{k}_{\mathrm{eff}}$ and $\boldsymbol{g}$ to achieve optimal performance. Alignment is critical for a single-axis

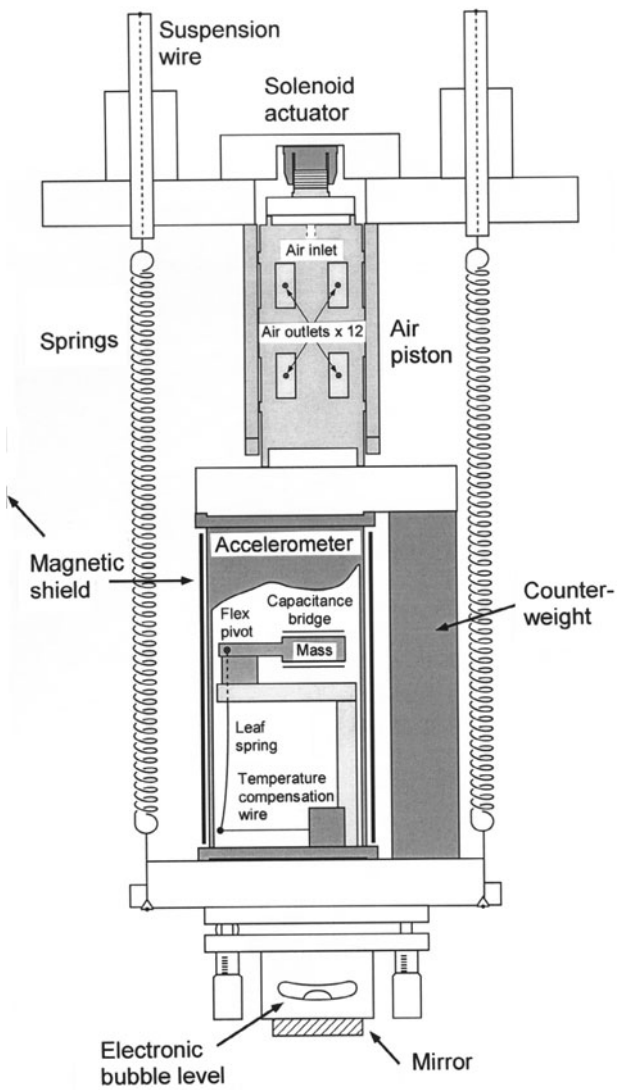

Figure 16. Mechanical system of the vibration isolator.

system such as this one. We have therefore expended considerable effort on making the axis of the air piston, the sensitive axis of the accelerometer and the direction of light propagation (normal to the mirror surface) parallel to each other and ultimately to the vector of gravity. To aid this alignment we have also added an electronic tilt sensor (Section 3.3.3). The accelerometer is offset to the side so that its internal test mass is on the symmetry axis of the system. This is important, since otherwise the system would translate benign rotational noise into unwanted vertical vibrations. To reduce its sensitivity to magnetic fields the accelerometer itself is fitted with a magnetic shield (Magnetic Shield Corp. 35P70) and additionally the whole vibration isolator is enclosed with magnetic-shielding material.

Figure 17 shows the performance of the vibration-isolation system. Without electronic feedback the combination of optical-table pneumatic isolators and vibration-isolator spring suspension already substantially reduces vibrations at frequencies above 3 Hz, but at the expense of some amplification at the mechanical resonance frequencies of 1.2 Hz (springs) and 1.8 Hz (table). Turning on the electronic feedback further reduces low vibrations in the 0.01 Hz to 10 Hz frequency band. The error signal (bold line) indicates a reduction by as much as a factor of 300, but without an independent sensor we cannot be sure that the true performance is as good. However, the great improvement of our interferometer signal provides strong supporting evidence.

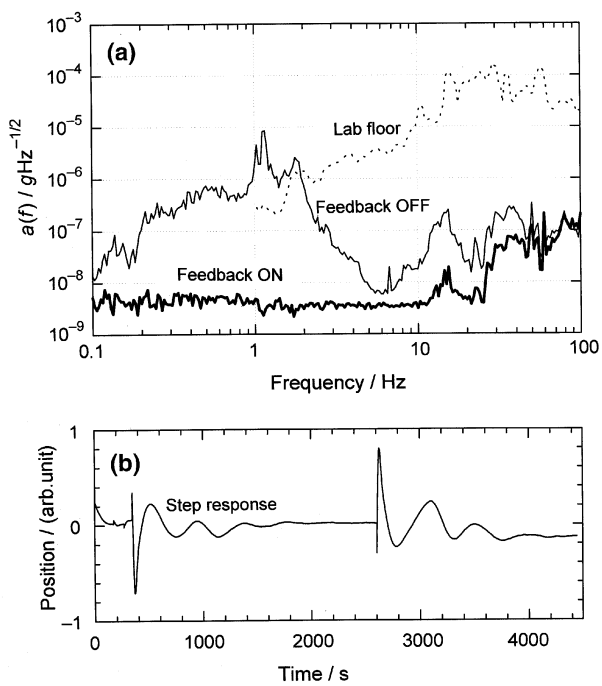

Figure 17. Performance of vibration isolator. (a) $a(f)$ is the acceleration noise spectral density measured by the vibration-isolator accelerometer. Passive spring isolation of the optical table and the isolator spring suspension (thin line) reduce the high-frequency vibrations relative to the laboratory floor (dotted line). Turning on the electronic feedback (thick line) further suppresses low-frequency vibrations. Below 10 Hz the error signal is smaller than the input noise of the spectrum analyser used for this measurement. (b) Position response to two steps in the force acting on the suspended mass. From $\omega_1 = \omega_0/\sqrt{G+1}$, we expect the effective resonance frequency to be ~0.01 Hz. However, we observe oscillation at ~0.0025 Hz, indicating that the low-frequency behaviour is not fully described by (24).

3.3.2 3-D active vibration isolation of optical table

While the main vibration-isolation system works well, its one-dimensional nature leads to intrinsic limitations on its performance. Most importantly, it cannot eliminate rotational vibrations. As the atom interferometer can also be sensitive to rotations (Sections 4.2.6, 5.5.1), these rotational vibrations, particularly when enhanced close to the resonance frequency (~2 Hz) of the table, could potentially dominate the interferometer noise.

To avoid this potential problem we have added a feedback system to control and vibrationally isolate the whole optical table. The system uses three single-axis, force-feedback accelerometers (Kinemetrics FBA-11, 10 V/g, flat acceleration response from dc to 90 Hz) to sense vertical accelerations at three different points of the L-shaped optical table (Figure 18). Their amplified signal is then digitized, electronically filtered and fed back to a set of three magnetic actuators (loudspeaker coils) mounted directly underneath each sensor. By reducing vertical vibrations at all three points, the system also suppresses rotations around the two orthogonal horizontal axes.

The feedback parameters of the system can be adjusted to achieve new table resonance periods of 3 s or 10 s. In either case the formerly troublesome

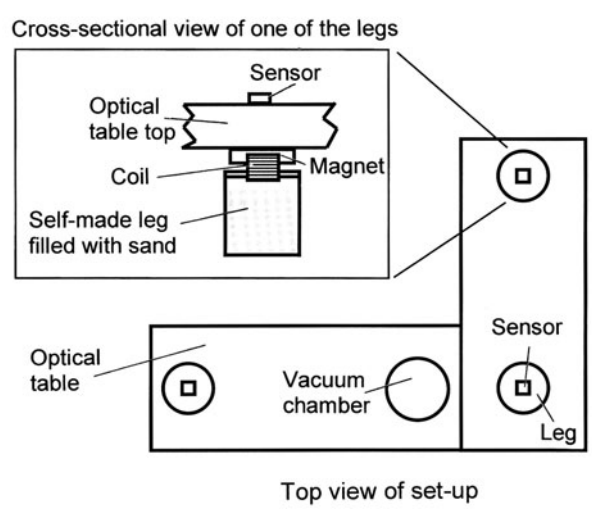

Figure 18. 3-D active vibration isolation and tilt control system.

resonances at 2 Hz are suppressed. Another important benefit of this system is the possibility of intentionally introducing rotations in a controlled way. This is essential for eliminating systematic errors arising from Coriolis effects (Section 5.5.1).

3.3.3 Tilt control

The surface of the retroreflection mirror has to be horizontal (Section 5.2.2) in order to achieve proper vertical alignment of the measurement axis. To achieve this, we have constructed a feedback system that controls the tilt of the optical table.

A high-resolution (0.1 μrad) electronic bubble level is attached to the back of the retroreflection mirror (Figure 16). Any wedge of the sensor with respect to the mirror surface can be measured by rotating the whole assembly through 360°. After performing this procedure we were able to determine the horizontal orientation of the mirror surface to within $\pm 5\,\mu\mathrm{rad}$.

The orientation of the set-up is controlled by pneumatic valves at each end of the L-shaped optical table. Air is allowed to leak out constantly, causing the table to tilt at $\sim 1000\,\mu\mathrm{rad/min}$. An electronic valve is used to adjust the inflow and control the tilt rate.

The feedback loop filter is set for a time constant of approximately 2 min. Since this is much longer than the time constant of the 3-D vibration-isolation system, the two systems do not disturb each other.

We tested the performance of the systems using a second, independent tilt sensor mounted to the main body of the optical table. The short-term regulation is better than 1 μrad. Over periods of several days there are very slow drifts of up to $\pm 20\,\mu\mathrm{rad}$, probably caused by slight warping of the tower structure supporting the vibration isolator and the main tilt sensors attached to it.

4. Experimental results

This section presents our experimental results, starting with the demonstration of interferometer fringes and proceeding to measurements of increasing complexity. We discuss the noise sources affecting these measurements and present the results of an experimental comparison between our atom interferometer and a classical "falling corner-cube" gravimeter.

Most of our measurements were performed using retroreflected Raman beams in $(lin \perp lin)$ configuration (Section 3.2.3) to drive Doppler-sensitive Raman transitions. The exceptions are a few tests conducted using Doppler-insensitive Raman transitions driven by co-propagating Raman beams in $(\sigma^+ - \sigma^+)$ configuration. This modification is useful since it allows one to investigate noise and systematic errors in a somewhat simpler system unaffected by vibrations or the initial velocity distribution of the atoms.

4.1 Interferometer fringes

Figure 19 shows a typical gravity fringe for an interferometer pulse spacing of $T = 160\,\mathrm{ms}$, limited by the finite size of the magnetically shielded measurement region. The fraction of atoms excited from the $F = 3$, $m_F = 0$ state to the $F = 4, m_F = 0$ state is plotted. The phase of the fringe is scanned by adjusting the frequency step size Δf (see Section 3.2.2). The Doppler-sensitive $\pi/2 - \pi - \pi/2$ interferometer sequence is implemented using square Raman pulses of equal intensity and durations of $40\,\mu\mathrm{s}$, $80\,\mu\mathrm{s}$, and $40\,\mu\mathrm{s}$.

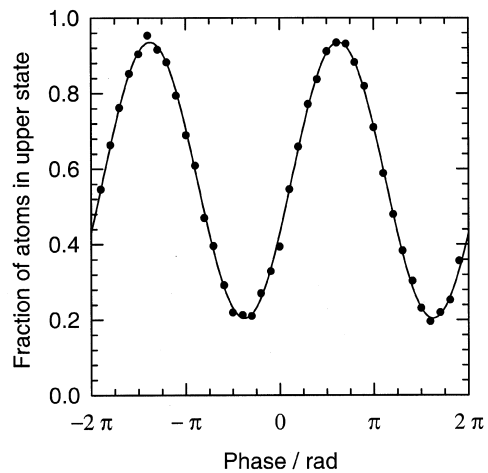

Figure 19. Typical Doppler-sensitive interferometer fringe for $T = 160\,\mathrm{ms}$. Each of the 40 data points represents a single launch of the atoms, spaced 1.3 s apart and taken over a period of 1 min. One full fringe period corresponds to $\sim 2 \times 10^{-6}\,g$. Performing a least-squares fit determines local gravity to approximately $3 \times 10^{-9}\,g$.

The fringe contrast (see (29) for definition) is 62 %, limited by the transfer efficiency of the Doppler-sensitive Raman pulses. It is almost independent of the interferometer pulse spacing T (67 % contrast for $T = 1\,\mathrm{ms}$).

Acquiring and fitting a sequence of such 1 min fringes allows us to measure gravity with a resolution of $2.0 \times 10^{-8}\,g/\sqrt{\mathrm{Hz}}$. Taking data by sitting on the slope of a fringe gives an even higher resolution of $1.5 \times 10^{-8}\,g$ for a single launch or $1.7 \times 10^{-8}\,g/\sqrt{\mathrm{Hz}}$.

4.2 Measurement noise

When we performed the first atom interferometric gravity measurements in 1991 [4], our measurement resolution of $\sim 3 \times 10^{-6}\,g$ per launch was limited by vibrations that caused phase noise in the Raman beams.

Our current system employs an active vibration-isolation system and has a resolution of $\sim 1.5 \times 10^{-8}\,g$ per launch. We now present a detailed discussion of the relevant noise sources and describe some of the methods that we have used to investigate their effects.

4.2.1 Definitions

We first define the term "resolution" as used in our experiment. Assume that we have a sequence of equally spaced gravity measurements $g_n = g(t_n)$ with $t_{n+1} = t_n + \tau$. One possible characterization of the measurement noise would be to use the ordinary variance

$$\sigma_g^2 = \left\langle (g_n - \langle g_n \rangle)^2 \right\rangle. \tag{25}$$

However, this variance generally depends on the number of samples in a finite sequence of gravity measurements and might diverge for infinite sequences. The two-sample variance

$$\sigma_g^2\,(2,\,\tau) = \frac{1}{2}\left\langle (g_{n+1} - g_n)^2 \right\rangle, \tag{26}$$

on the other hand, does not depend on the number of samples. We therefore refer to this quantity as the resolution of the gravity measurement.

It is also possible to define the two-sample variance $\sigma_g^2(2,t)$ for longer time separations $t = N\tau$ by first taking the mean of N consecutive measurements. It is given by

$$\sigma_g^2\,(2,\,t) = \frac{1}{2}\left\langle \left(\sum_{k=0}^{N-1} \frac{g_{n+k}}{N} - \sum_{k=N}^{2N-1} \frac{g_{n+k}}{N} \right)^2 \right\rangle. \tag{27}$$

If the noise of consecutive measurements g_n is uncorrelated, then we have

$$\sigma_g\,(2,\,t) = \frac{1}{\sqrt{N}}\,\sigma_g\,(2,\,\tau)$$
$$\text{and} \quad \sigma_g\,(2,\,\tau) = \sigma_g. \tag{28}$$

As this is the case for most of the noise sources that we have considered (except for very-low-frequency vibrations and rotations), we often do not distinguish between $\sigma_g(2,\tau)$ and σ_g.

We can also relate the variance of g to the signal-to-noise ratio of the experimentally observed interferometer fringes. Assume that the number of detected atoms is given by

$$x = A\,[1 + C\,\cos\,(\Delta\phi)], \tag{29}$$

where $\Delta\phi = k_{\text{eff}} g T^2$ is the interferometer phase, C the fringe contrast and AC the observed fringe amplitude. We then define the signal-to-noise ratio $(S/N) = AC/\sigma_x$ by comparing the variance σ_x of the signal on the slope of a fringe with the fringe amplitude AC. Assuming that neither A nor C fluctuate, the S/N ratio and the variances of interferometer phase and gravity value are related by

$$(S/N)^{-1} = \sigma_{\Delta\phi} = k_{\text{eff}} T^2 \sigma_g. \tag{30}$$

When theoretically evaluating the effect of potential noise sources, we generally assume $100\,\%$ fringe contrast.

4.2.2 Shot noise and detection noise

The atomic shot noise would ultimately limit the resolution of the interferometer at a level of $\Delta g/g \propto 1/\sqrt{N}$, where N is the number of detected atoms. In our experiment $\sim 3 \times 10^6$ atoms contribute to the interferometer signal, corresponding to a shot-noise-limited S/N ratio of $1700/1$.

One prerequisite for shot-noise-limited performance is a detection efficiency sufficiently close to unity. Our fluorescence-detection scheme fulfills this condition, since ~ 5 photo-electrons are produced per atom. However, the detection S/N ratio is only $300/1$, limited by frequency fluctuations of the intensity-stabilized probe beam.

4.2.3 Raman laser intensity noise

The Raman beam intensities are not actively stabilized and have intensity fluctuations with a variance of $\sigma_I/I < 0.5\,\%$ in the frequency range $1\,\text{Hz}$ to $1\,\text{kHz}$. After the first $\pi/2$ pulse the variance of the atomic inversion W ($W = [x - A]/A$ in (29), we assume $C = 1$) is

$$\sigma_W = \frac{\pi}{2}\,\frac{\sigma_\Omega}{\Omega} = \frac{\pi}{2}\,\frac{\sigma_I}{I}. \tag{31}$$

After the complete interferometer sequence, and assuming that the intensity fluctuations for all three pulses are uncorrelated, the variance of W on the slope of a fringe is estimated to be

$$\sigma_W = \frac{\sqrt{3}\pi}{2}\,\frac{\sigma_\Omega}{\Omega} = \frac{\sqrt{3}\pi}{2}\,\frac{\sigma_I}{I}. \tag{32}$$

Using the relation $(S/N)^{-1} = \sigma_W$, which is valid on the slope of a fringe, we find that the $0.5\,\%$ intensity fluctuations correspond to a S/N ratio of $75/1$.

4.2.4 Raman laser phase noise

All the interferometer phase shifts are measured relative to the phase of the Raman laser beams and any Raman phase noise therefore translates directly into noise of the gravity measurement. According to $\Delta\phi = k_{\text{eff}} g T^2$, the same amount of phase noise corresponds to smaller amounts of acceleration noise at longer interferometer pulse spacings T. There are three basic contributions to the Raman phase noise.

Noise introduced by phase-lock system. Noise arising from the phase-locking system is small when the two Raman beams are first sent through an optical fibre together before their beat note is detected by the fast photodiode. If the photodiode is placed before the fibre the Raman beams are not *perfectly* overlapped when part of the light is split off and sent to the photodiode. This small imperfection causes the Raman

phase at the photodiode (controlled by the phase lock) to differ from the one at the fibre coupler (seen by the atoms). Air currents and slight piezo movements in the external-cavity lasers (Littrow configuration) affect this difference and cause noticeable amounts of phase noise ($\sim 0.05\,\mathrm{rad}$ root-mean-square when integrated from 1 Hz to 100 Hz.

Short-term stability of frequency reference. The short-term stability of the quartz frequency reference, usually expressed in terms of the Allan variance $\sigma_{\mathrm{Allan}}(T)$ (or more precisely, $\sigma_y(2,T,T,f_{\mathrm{c}})$, the two-sample variance of the fractional frequency, without dead time, bandwidth limited at f_{c} [34]). In fact, in the absence of other noise sources the atom interferometer phase noise is related to the Allan variance by the formula

$$\sigma^2_{\mathrm{Allan}}(T) = \frac{1}{2\,(2\,\pi\,f_0\,T)^2}\,\sigma^2_{\Delta\phi}, \tag{33}$$

where T is the time between interferometer pulses and f_0 the Raman frequency difference of $\sim 9.2\,\mathrm{GHz}$ [23]. The Allan variance depends on the measurement bandwidth f_{c}, which in this case is approximately $1/(2\tau)$, where τ is the length of a single interferometer pulse.

We have measured the Allan variance of several quartz oscillators by comparing them. The measured Allan variance of our Loran-C frequency reference at 1 s is $\sigma_{\mathrm{Allan}}(1\,\mathrm{s}) = 8\times 10^{-13}$, an improvement of approximately a factor of 6 on the product specification following our work on their electronics [9]. Assuming that the Allan variance at $0.16\,\mathrm{s}$ is similar, it would only limit our measurement S/N ratio to $95/1$.

High-frequency phase noise. White phase noise of our microwave frequency source has been identified as the greatest problem. It is dominated by the phase noise of our $10\,\mathrm{MHz}$ reference when multiplied up to microwave frequencies (frequency multiplication increases phase noise by $20\,\mathrm{dB}$ for every multiplication by 10). The measured white phase noise of the Loran-C frequency reference when multiplied up to $9.18\,\mathrm{GHz}$ is $S_\varphi = 2\times 10^{-4}\,\mathrm{rad}/\sqrt{\mathrm{Hz}}$. Our frequency-multiplier chain provides no filtering to reduce this noise, as it was originally intended for use in atomic clocks where this noise was not the dominant problem. In this respect, a good microwave synthesizer actually offers better performance.

High-frequency phase noise affects the interferometer measurement by causing uncorrelated phase fluctuations for each of the interferometer pulses. The variance $\sigma_\phi \simeq S_\varphi/\sqrt{2\tau}$ of these fluctuations depends on the pulse length τ, which determines the cut-off frequency f_{c} for integration of the white phase noise. It has thus slightly different values, $\sigma_\phi(\pi)$ and $\sigma_\phi(\pi/2)$, for different interferometer pulses. On the slope of an interferometer fringe the variance of the total interferometer phase is then given by

$$\begin{aligned}\sigma_{\Delta\phi} &= \sqrt{\sigma_\phi\,(\pi/2)^2 + 4\,\sigma_\phi(\pi)^2 + \sigma_\phi\,(\pi/2)^2}\\ &\approx 2\,S_\varphi/\sqrt{\tau_\pi}.\end{aligned} \tag{34}$$

For π pulse lengths of $100\,\mu\mathrm{s}$ this would limit our S/N ratio to $25/1$. This noise contribution probably limits the performance of the interferometer when operated in Doppler-insensitive mode. Replacing the frequency-multiplier chain with a borrowed microwave synthesizer (HP-8665-A), we have observed somewhat lower noise levels. The high-frequency phase noise of this synthesizer should allow a S/N ratio of $\sim 100/1$. However, other noise sources start to dominate much earlier.

4.2.5 Vibrations

As our set-up uses retroreflected Raman beams, only (vertical) vibrations of the retroreflection mirror should be relevant. Let us initially assume that these vibrations have a fixed frequency and a well-defined phase. We can then write the vibrational acceleration as

$$\boldsymbol{a}\,(t) = \widehat{\boldsymbol{a}}\,(a_{\mathrm{c}}\,\cos\omega t + a_{\mathrm{s}}\,\sin\omega t), \tag{35}$$

where the time $t=0$ coincides with the central π pulse of the interferometer sequence ($\hat{\boldsymbol{a}}, \hat{\boldsymbol{v}}_{\mathrm{rec}}, \ldots$ are unit vectors). There are several ways to derive the resulting interferometer phase shift, for example, using a perturbative treatment [22, 19] and the perturbation Lagrangian $\mathcal{L} = m\,\boldsymbol{a}\cdot\boldsymbol{r}$. For reasons of symmetry the result is not affected by the cosine component in (35). The sine component causes a shift of

$$\Delta g = \frac{4\,a_{\mathrm{s}}}{T^2}\,\widehat{\boldsymbol{a}}\cdot\widehat{\boldsymbol{v}}_{\mathrm{rec}}\,\frac{\sin^2\,(\omega T/2\,)}{\omega^2}, \tag{36}$$

which clearly depends on the vibration frequency as well as the interferometer pulse spacing (Figure 20a). In fact, for each pulse separation T there are certain frequencies at which vibrations have no effect. This occurs whenever an integer number of vibrational cycles fits into the interval between π pulses and $\pi/2$ pulses. Another important property is the second-order roll-off at frequencies higher than $1/T$, which means that the interferometer effectively low-pass filters the acceleration signal. Figure 20b shows the vibration-isolator error signal before and after implementing the improvements illustrated in Figure 21, and Figure 20c shows the noise arising from these vibrations computed with (36).

We can also use the above formula to calculate the noise of the interferometer signal (for a specific T) from the power spectrum of residual accelerations at the vibration isolator. One problem with this power spectrum is that the real accelerations in the lowest frequency decade are obscured by the digitization noise floor. We estimate the measurement noise arising from residual vibrations to be 5 μGal, corresponding to S/N ratios of $55/1$.

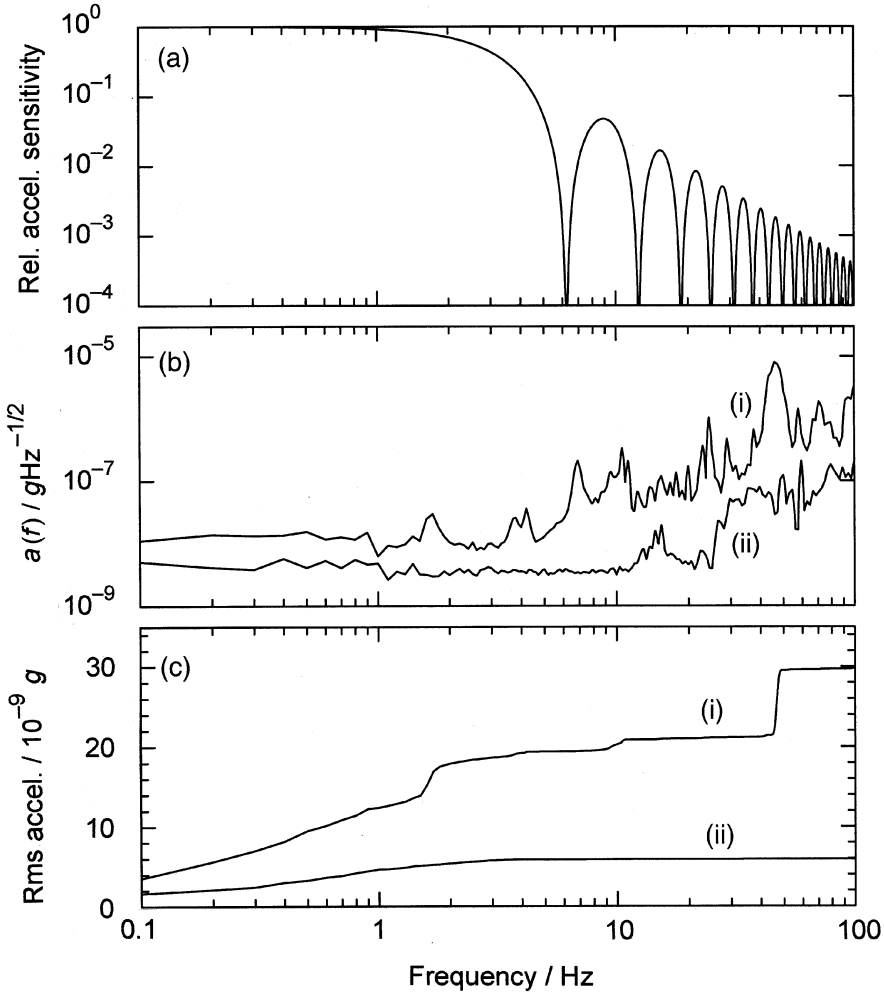

Figure 20. Effect of vibrations. (a) Relative sensitivity to different vibrational frequencies for an interferometer pulse spacing of $T = 160\,\mathrm{ms}$. (b) Vibration-isolator error signal: $a(f)$ is the acceleration noise spectral density (i) before and (ii) after implementing improvements. (c) Predicted noise of interferometer signal due to vibrations (weighted) integrated up to a certain frequency (i) before and (ii) after the improvements.

4.2.6 Rotations

The interferometer can be sensitive to rotations as well as to vibrations. Besides the systematic errors arising from the Earth's rotation, this sensitivity can also cause noise problems. The level of rotational noise is largely determined by the performance of the optical table, which reduces rotations at high frequencies but actually enhances them close to its resonance frequency of $\sim 2\,\mathrm{Hz}$. The main active vibration isolator suppresses the $\sim 2\,\mathrm{Hz}$ vibrational resonance but does nothing to suppress rotations. This is one of the reasons why we added an active 3-D vibration-isolation system for the optical table. While this system thoroughly eliminates problems at the original table resonance, it currently does nothing at frequencies above $10\,\mathrm{Hz}$.

As the main vibration isolator performs very well over a wide frequency range, we can assume that it eliminates all vertical movement at the sensor location. In effect, we therefore only have to consider rotations around this point. We first consider rotational noise with a well-determined frequency ω and phase which can be written as

$$\boldsymbol{\Omega}(t) = \widehat{\boldsymbol{\Omega}}(\Omega_\mathrm{c} \cos \omega t + \Omega_\mathrm{s} \sin \omega t), \tag{37}$$

where the time $t = 0$ coincides with the central π pulse of the interferometer sequence. As in the case of vibrations, we can derive the resulting interferometer phase shift using a perturbative treatment and the perturbation Lagrangian $\mathcal{L} = m\boldsymbol{\Omega} \cdot (\boldsymbol{r} \times \boldsymbol{v}) + \frac{1}{2} m (\boldsymbol{\Omega} \times \boldsymbol{r})^2$. The sine and cosine components in (37) give rise to different shifts in the measured acceleration:

$$\Delta g_\mathrm{c} = \frac{2\Omega_\mathrm{c}}{T} \widehat{\boldsymbol{\Omega}} \cdot (\boldsymbol{v}_0 \times \widehat{\boldsymbol{v}}_\mathrm{rec}) \frac{\sin(\omega T)}{\omega} \tag{38}$$

$$\Delta g_\mathrm{s} = -\frac{4\Omega_\mathrm{s}}{T^2} \widehat{\boldsymbol{\Omega}} \cdot (\boldsymbol{x}_0 \times \widehat{\boldsymbol{v}}_\mathrm{rec}) \frac{\sin^2(\omega T/2)}{\omega}. \tag{39}$$

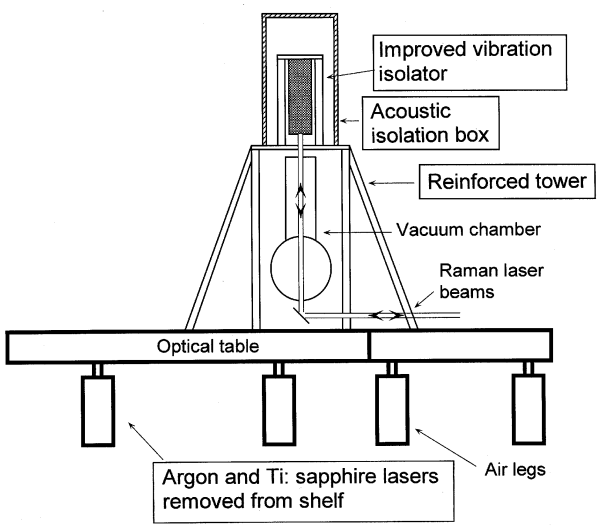

Figure 21. Evolution of the experimental set-up to reduce noise levels. The boxed labels refer to improvements that reduce the noise level.

Here, $\boldsymbol{x}_0$ and $\boldsymbol{v}_0$ are the atomic position (relative to the sensor in the vibration isolator) and velocity (taking the mean of both interferometer paths) at the time of the central interferometer π pulse. There are several ways to derive these formulas, for example by using a perturbative treatment and the perturbation Lagrangian $\mathcal{L} = m\boldsymbol{\Omega}\cdot(\boldsymbol{r}\times\boldsymbol{v}) + \frac{1}{2}m(\boldsymbol{\Omega}\times\boldsymbol{r})^2$. We have neglected the centrifugal force terms as they are small relative to the other contributions.

Note that while for constant rotation rates ($\omega = 0$) the first contribution reduces to the familiar Coriolis force term, $\Delta g_{\mathrm{s}} = \boldsymbol{\Omega}\cdot(\boldsymbol{v}_0\times\widehat{\boldsymbol{v}}_{\mathrm{rec}})$, the second contribution will vanish.

Both effects become relevant when the experimental apparatus is misaligned in one way or another. The first effect is proportional to the horizontal velocity $\delta v = \boldsymbol{v}_0\cdot\widehat{\boldsymbol{v}}_{\mathrm{rec}}$, the second to the horizontal offset $\delta x = \boldsymbol{x}_0\cdot\widehat{\boldsymbol{v}}_{\mathrm{rec}}$ between atoms and sensitive point of the vibration isolator ($\widehat{\boldsymbol{v}}_{\mathrm{rec}}$ is vertical). The horizontal velocity δv is generally very small ($< 2\times10^{-4}\,\mathrm{m/s}$), as it is carefully adjusted in order to avoid systematic errors caused by the Earth's rotation. The horizontal displacement is harder to quantify and probably substantial. We have actually been able to reduce the measurement noise by horizontally translating the vibration isolator in increments of $5\,\mathrm{mm}$, so $5\,\mathrm{mm}$ should be a good estimate for δx.

We have measured the rotational noise of the optical table and used the above formulas with explicit values for δx and δv to predict the resulting interferometer noise. We found that the effect proportional to δv is almost certainly irrelevant, even for an unrealistically large horizontal velocity of $1\,\mathrm{mm/s}$. The effect proportional to δx, however, can be more substantial, especially if the active feedback is turned off. In that case it might cause fluctuations of up to 3 μGal, corresponding to a S/N ratio of $90/1$.

4.2.7 Measured noise

The noise observed during an interferometer measurement depends on the exact operating parameters such as pulse separation and pulse length. Varying these parameters for both Doppler-sensitive and Doppler-insensitive interferometer configurations provides useful information for disentangling the effects of the various noise sources such as vibrations, phase noise, etc.

Our standard method of determining the S/N ratio is to sit on the slope of an interferometer fringe and determine the variance within sets of 40 consecutive launches. While this variance could potentially depend on the number of launches (see Section 4.2.1), we find that this is not the case for our experiment and for fewer than ~100 launches (see Figure 22a).

Calculating the two-sample variance $\sigma_g(2,t)$ defined in (27) (or $\sigma_{\Delta\phi}(2,t)$ for Doppler-insensitive fringes) of a long sequence of measurements provides additional information. Figure 22 shows these variances for a variety of measurement conditions.

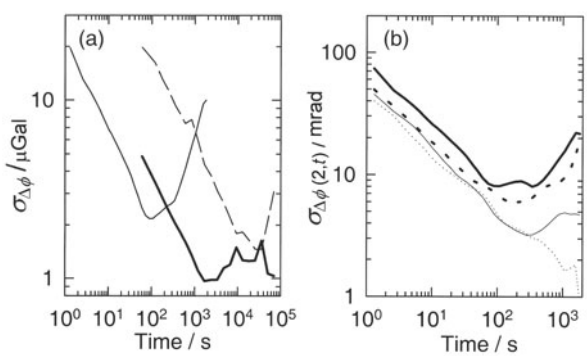

Figure 22. Two-sample variance of interferometer measurements. (a) For Doppler-sensitive configuration with pulse separation $T = 160\,\mathrm{ms}$ and data taken either by sitting at the side of a fringe (thin line) or by fitting 1 min gravity fringes (thick line). The equivalent one for the FG5 gravimeter (dashed line, see below) is also given. (b) Comparison between Doppler-sensitive (thick lines) and insensitive (thin lines) configurations for pulse spacings of $T = 160\,\mathrm{ms}$ (solid) and $T = 2\,\mathrm{ms}$ (dotted).

One obvious feature is the $1/\sqrt{t}$ dependence for fewer than ~100 launches. This result justifies our previous definition of the S/N ratio, as it shows that for sets of fewer than 100 launches, noise of different launches is uncorrelated. In this case we have $(S/N)^{-1} = \sigma_{\Delta\phi}(t_1)$, where $\Delta t_1 = 1.3\,\mathrm{s}$ is the time between launches. It also means that all the noise can be attributed either to high-frequency sources, with periods shorter than the $1.3\,\mathrm{s}$ interval between launches, or to very-low-frequency effects with typical timescales of several minutes or more. No dominant noise sources seem to exist with periods between 1 s and 100 s.

Our main reason for calculating the two-sample variance was to determine up to what time the $1/\sqrt{t}$ dependence holds and to find the eventual flicker-noise

floor. For data taken by sitting on the slope of a fringe (thin line in Figure 22a) the increase of the noise around $t = 100\,\mathrm{s}$ is probably caused by drifts (mostly PMT dark current) in the detection system.

The method of taking data by fitting full interferometer fringes is not vulnerable to such drifts and therefore better suited for long-term measurements. However, since data points at the extrema of a fringe provide no phase information, and because of experimental overhead, this method has slightly lower resolution at short times.

The thick line in Figure 22a is for one of our multi-day gravity datasets which was acquired in this way and corrected for tides using a model that includes ocean-loading effects (see Section 4.3). The increase in noise around $t = 2 \times 10^3\,\mathrm{s}$ is probably due to residual tidal effects. A diurnal ($\sim$12 hour period) component with an amplitude of 4.5 μGal (compare Figure 24) would explain the peak at $t = 2 \times 10^4\,\mathrm{s}$.

The high-frequency noise, which affects every single launch in the same way, shows very little dependence on the time between pulses and is also not very different for Doppler-sensitive and Doppler-insensitive modes. The only times we have seen substantially lower noise levels are when we either replaced our frequency-multiplier chain with a microwave synthesizer (HP-8665-A) for a short time between pulses ($T \sim 1\,\mathrm{ms}$) or used much longer pulse lengths (up to $5\,\mathrm{ms}$, only possible in Doppler-free mode). However, the microwave synthesizer does not significantly improve the noise for the 160 ms pulse separation Doppler-sensitive fringes that we use for our gravity measurements.

4.2.8 Summary

Table 3 summarizes the effects of the various noise sources. We are able to account for most of the observed noise. At present, the noise is 19 parts in 10^9 for each launch of 1.3 s, more than a factor of 100 away from the atom shot-noise limit.

Table 3. Known noise sources and their estimated effect on a gravity measurement with $T = 160\,\mathrm{ms}$ between interferometer pulses of lengths $\tau_\pi = 80\,\mu\mathrm{s}$ and $\tau_{\pi/2} = 40\,\mu\mathrm{s}$. The noise terms assume the interferometer fringe contrast to be 100 %. For our observed contrast of $\sim$66 %, the amplitude noise terms (the first two) should be multiplied by $3/2$.

Noise source	S/N	σ_g/μGal
Atom shot noise (3×10^6 atoms)	1700	0.16
Detection noise	300	0.9
Loran-C frequency stability	95	3.0
Raman laser intensity noise	75	3.5
Residual vibrations and rotations	55	5
Background fluctuations	40	7
High-frequency phase noise	25	11
Overall known noise sources	**18**	**15**
Observed noise	**14**	**19**

4.3 Long-term gravity data

To perform useful measurements any gravimeter has to be able to operate continuously over extended periods of time. Gravity data taken over a 9-day period (Figure 23) show a complicated time dependence caused mainly by tidal forces. Although these tidal effects are often an object of study themselves, they are considered to be a predictable perturbation in the context of absolute gravity measurements and can be accounted for with semi-empirical tide models.

Since the tidal effects are large compared with the instrumental resolution and targeted accuracy, it is essential to model them properly. At first glance the theoretical curve in Figure 23 seems to match the experimental data extremely well. However, detailed examination of the data in Figure 24 makes clear that standard tide models, even if they include the Earth's elastic response to tidal forces, are inadequate at this level of precision.

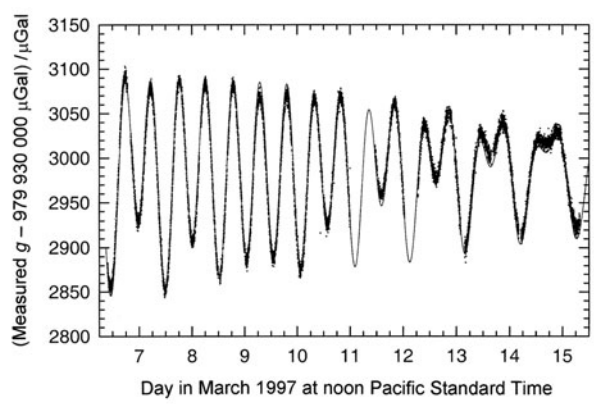

Figure 23. Gravity data taken over a period of 9 days. The data points are spaced approximately 1 min apart and each corresponds to one of the interferometer fringes shown in Figure 19. The data clearly show the complex variations in g caused (mostly) by tides. The solid line is a single parameter (vertical offset) fitted using a theoretical model of the gravity tides at our measurement site (Stanford, California).

The obvious discrepancy between the standard tide model and the experimental data can be attributed to a phenomenon known as ocean loading [35, 36]. Ocean loading is caused by the redistribution of the water in the oceans in response to tidal forces. When more water is present at a certain location, its mass will increase the pressure on the ocean floor and depress it slightly, typically by a few centimetres. This depression of our local tectonic plate, not direct gravitational attraction by the water, is the cause of the periodic residuals in Figure 24b. We can remove most of these periodic residuals by including an ocean-loading model [37, 38].

4.4 Comparison with classical gravimeter

We have compared the value of g measured by our atom interferometer against one obtained by a gravimeter

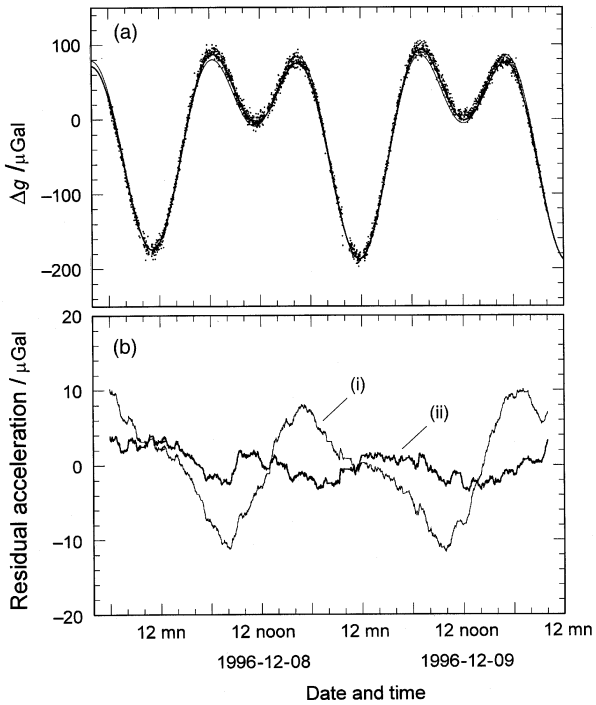

Figure 24. (a) A closer look at two days of gravity data. Each data point represents a 1 min gravity measurement. The solid lines represent two different tidal models. (b) The residuals of the data with respect to a tidal model where (i) the Earth is modelled as a solid elastic object and (ii) the effects of ocean loading of the Earth are taken into account. Effects at the level of a few parts in 10^9 and changes in the local barometric pressure have not been included.

based on an optical Michelson interferometer with a freely falling corner-cube in one arm.

Since g is time-varying and site-dependent, we performed this comparison over a 3-day period and in our own laboratory, using an FG5 absolute gravimeter [15] operated by the National Oceanic and Atmospheric Administration (NOAA). The accuracy of this instrument is usually specified as 2 μGal, but data from international gravity comparison measurements [39] indicate that 4 μGal is a more realistic value.

In addition to testing the accuracy of the atom interferometer we were interested in comparing the measurement noise of the two instruments. The resolution of the FG5 gravimeter is specified to be as good as 5 μGal for a single drop at a seismologically quiet site. With a maximum repetition rate of 1 drop per second this would mean a resolution of 5 $\mu\text{Gal}/\sqrt{\text{Hz}}$, substantially better than the 20 $\mu\text{Gal}/\sqrt{\text{Hz}}$ achieved by our atom interferometer. However, our laboratory is definitely not a seismologically quiet site. It is located on the second floor of the busy physics building which itself is surrounded by major demolition and construction sites.

The preferred drop rate of the FG5 is 1 drop every 15 s. Higher duty cycles would reduce the lifetime of the instrument and the lack of sufficient settling time for drop-related vibrations could compromise its accuracy. The repetition rate of our atom interferometer, on the other hand, is basically determined by the interferometer pulse spacing T and the loading time for the MOT. Currently it is limited to one launch every 1.3 s by the time needed to reprogram the DDS synthesizer between launches.

Figure 25 summarizes the original data from comparison measurements. In order to compare the resolution of the two instruments with their different sampling rates, we binned the data into same-length intervals (1 min, corresponding to 40 atom launches and 4 corner-cube drops). The noise of the atom interferometer measurement is lower by a factor of 4 when normalized to measurement time, and comparable to that of the other instrument on a per drop basis.

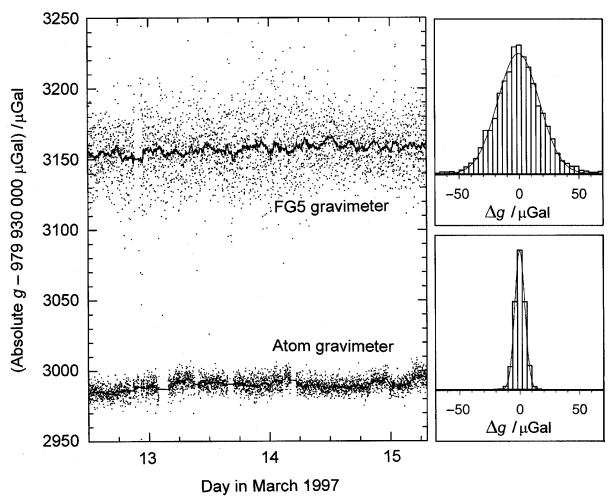

Figure 25. Comparison between atom interferometer (lower data set) and classical gravimeter (upper data set). Each point represents 1 min of tide-corrected gravity data. The continuous lines are running averages (30 min). The histograms show distribution of the 1 min data points around the mean (for every 30 min interval). Compared with the classical instrument, the measurement noise of the atom interferometer is smaller by a factor of 4 for equivalent measurement times. The noise per drop (launch), on the other hand, is comparable. Both instruments observe a slight increase in g over the measurement period owing to changes in atmospheric pressure (see Section 5.6).

Comparing the absolute gravity values is more complicated as the measurements were taken at slightly different locations and measurement heights (Section 5.5.2). After accounting for the different locations, we found that there was still an offset of approximately 175 μGal (Figure 25). We later discovered that most of this discrepancy could be attributed to an incorrect published value for the caesium D_2 wavelength (see Section 5.2.1). The use of a better wavelength value to analyse our data immediately reduced the discrepancy to less than 40 μGal. Since the time of the simultaneous run we have identified several other systematic effects, which are discussed in the next section.

5. Systematic effects

An important property of the atomic fountain gravimeter is its ability to obtain an absolute gravity value directly referenced to atomic standards. However, the ultimate accuracy of the measurement still depends on our understanding of potential systematic errors.

Gravity measurements differ from those of fundamental constants in that g is a site-dependent quantity that also varies with time and is affected by environmental conditions, particularly tidal effects. Models of these effects are therefore needed in order to determine the final measurement precision and accuracy.

This section presents our analysis of systematic effects. We mostly concentrate on effects that could cause gravity offsets of more than 0.1 μGal and comment only briefly on the smaller effects that we have nevertheless considered. Related discussions of systematic errors for atom interferometers and atomic clocks can be found in [16, 34].

5.1 Displacements of the atomic transition frequency

5.1.1 AC Stark shifts

AC Stark shifts have to be considered for light-pulse atom interferometers. In the case of 2-photon, off-resonant Raman transitions they are intrinsically related to the Rabi frequency and cannot be avoided. Figure 26 illustrates the frequency shifts $\Omega_{\mathrm{a}}^{\mathrm{ac}}$ and $\Omega_{\mathrm{b}}^{\mathrm{ac}}$ of the two ground states caused by the presence of the Raman light fields.

The average ac Stark shift $\Omega_{\mathrm{avg}}^{\mathrm{ac}} = (\Omega_{\mathrm{a}}^{\mathrm{ac}} + \Omega_{\mathrm{b}}^{\mathrm{ac}})/2$ is the same for atoms in both paths of the interferometer, as long as one can neglect the intensity variations of the Raman beams over the path separation. This component of the ac Stark shift therefore leads to no observable phase shift.

The differential ac Stark shift $\delta^{\mathrm{ac}} = \Omega_{\mathrm{a}}^{\mathrm{ac}} - \Omega_{\mathrm{b}}^{\mathrm{ac}}$, on the other hand, can lead to an observable phase shift. Combining the results for π and $\pi/2$ pulses from [16], we find that the resulting first-order phase shift for a Mach-Zehnder type interferometer is given by

$$\Delta\phi^{\mathrm{ac}} = \frac{\delta_3^{\mathrm{ac}}}{\Omega_{\mathrm{eff}}} - \frac{\delta_1^{\mathrm{ac}}}{\Omega_{\mathrm{eff}}}, \tag{40}$$

where Ω_{eff} is the effective 2-photon Rabi frequency and δ_1^{ac} and δ_3^{ac} are the Stark shifts during the first and last $\pi/2$ pulse, respectively. Thus, there is a strong cancellation of ac Stark shift effects as the Raman beam parameters are the same for both pulses. However, the spreading of the atomic cloud with respect to the Gaussian Raman laser intensity profile breaks the symmetry and causes a substantial residual shift.

This residual shift can be essentially eliminated by properly adjusting the detuning and intensity of the Raman beams. For our experimental detuning of $\Delta \approx 1\,\mathrm{GHz}$ from the intermediate states, state $|a\rangle$ is mostly shifted by the Raman beam with frequency ω_1 and state $|b\rangle$ by the beam with frequency ω_2. The contribution from the other beams is much smaller owing to much greater detuning of $\sim 8\,\mathrm{GHz}$ (see Figure 26). Therefore, the ac Stark shifts can be adjusted almost independently and made the same by changing the intensity of the relevant Raman beam.

We have also determined the correct intensity ratio using an experimental method (Figure 27). An additional Raman pulse of variable length is added to the interferometer sequence. It is substantially detuned ($\sim 1\,\mathrm{MHz}$) from the Raman resonance so that it does not actually drive a Raman transition. On the other hand, this detuning is small compared with the 1-photon detuning Δ and the ac Stark shifts are therefore the same as during normal interferometer pulses. If the intensities of the beams are not properly balanced, then the additional pulses cause an interferometer phase shift

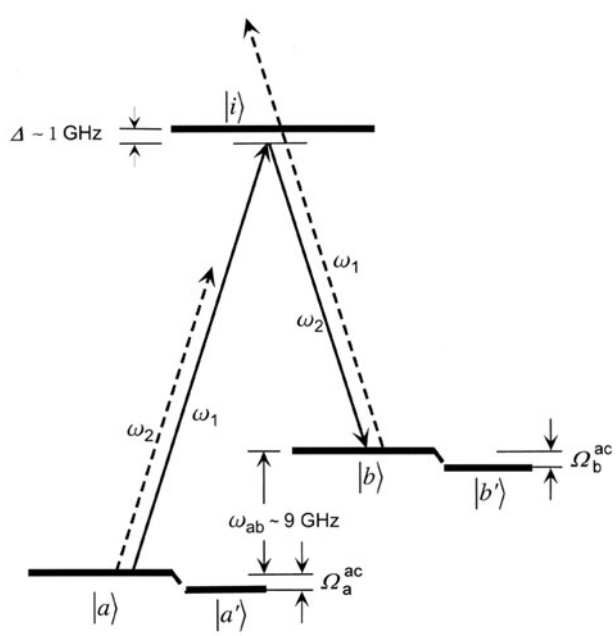

Figure 26. Level scheme for stimulated Raman transitions.

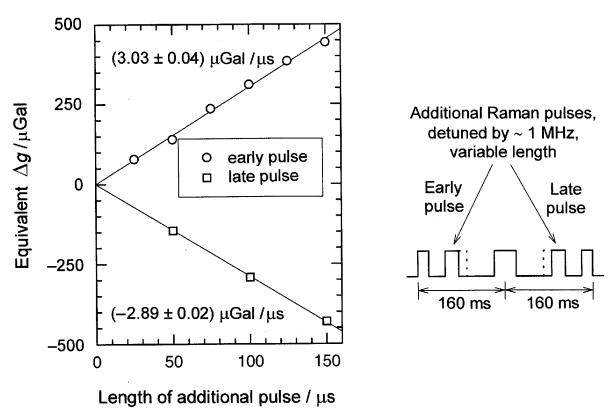

Figure 27. Measurement of the differential ac Stark shift caused by imbalanced Raman beams.

which is proportional to the pulse length. The sign of the shift is different for pulses introduced during transit in the first and second half of the interferometer.

The results shown are for a detuning of $\Delta = 0.99$ GHz from the $F' = 4$ excited state and for $I_{F=4}/I_{F=3} = 2$ ($I_{F=4}$ and $I_{F=3}$ are the intensities of the Raman lasers coupling primarily to the $F = 4$ and $F = 3$ ground states, respectively). One can now eliminate the differential ac Stark shift by adjusting the intensity ratio until the phase shifts caused by the additional pulses disappear ($I_{F=4}/I_{F=3} = 1.67$ for $\Delta = 0.99\,\mathrm{GHz}$). Note that the slope of the data in Figure 27 can be used to accurately measure ac Stark shift. As a comparison see the work of Müller et al. [40].

Figure 28a shows that the experimentally determined intensity ratio is a function of the detuning Δ, in good agreement with the theoretical result obtained by calculating the actual shifts due to the coupling of both beams to each hyperfine level of the excited state.

The Raman laser intensities are not actively stabilized and can change over the course of an experiment. Even after the differential ac Stark shift has been eliminated by balancing the Raman beams, it is therefore still important to know the residual sensitivity to small changes in the intensity ratio. Figure 28b shows that for our standard experimental configuration the measured gravity value is rather insensitive to changes in the intensity ratio (which typically stays between 1.65 and 1.75). The associated uncertainty is ± 1 μGal. To achieve sub-microgal resolution it would be necessary to either reduce the sensitivity even further or to implement active intensity stabilization.

5.1.2 Zeeman shifts

In our interferometer the atoms are always kept in magnetic-field-insensitive states with $m_F = 0$.

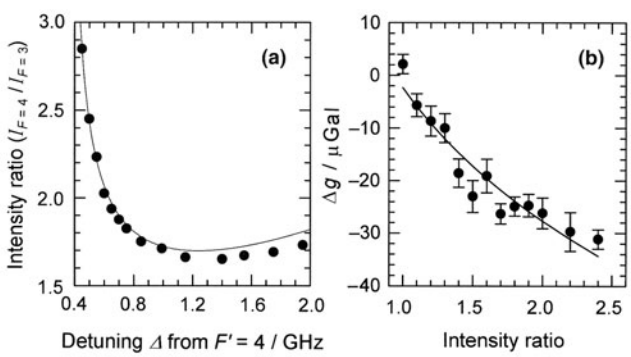

Figure 28. (a) Detuning dependence of the Raman laser intensity ratio at which the differential ac Stark shift vanishes. The data points are obtained using the method from Figure 27, the solid line is the theoretical model described in the text. (b) Residual sensitivity of the gravity measurement to small changes in the Raman laser intensity ratio, for a pulse separation of $T = 160\,\mathrm{ms}$ and a 1-photon detuning $\Delta = 0.99\,\mathrm{GHz}$ from the $F' = 4$ excited state. The solid line is a two-parameter (offset, amplitude) fit using the theoretical model.

However, these states still show a quadratic Zeeman shift and for small magnetic fields the resulting change of the caesium hyperfine frequency is given by

$$\Delta f_{\mathrm{hfs}} = 427.45\,\mathrm{Hz/G^2}. \tag{41}$$

This effect is large enough to require well-controlled magnetic fields and extensive magnetic shielding to achieve the millihertz frequency stability necessary for gravity measurements at the 1 μGal level. As the effect is quadratic in the magnetic field, it is desirable to operate at the smallest possible bias field. However, a certain bias-field strength is necessary to define a proper quantization axis and to prevent Majorana spin flips when the atoms enter the magnetically shielded region. Preferably, the field should also be big enough to separate the frequencies of magnetic-field-sensitive transitions from the magnetic-field-insensitive ones. As we have to use relatively short Raman pulses (80 μs π pulse, 40 μs π/2 pulse) to address a reasonably wide atomic velocity distribution, this would require a bias field of at least 0.15 G.

In principle, it would be possible to estimate the effect of residual magnetic-field inhomogeneities theoretically. But since the magnetic field distribution inside the magnetic shield is relatively complex (see Figure 29), experimental tests are easier and more conclusive. The results in Figure 30a indicate that the measured gravity value changes substantially at strong bias fields. This makes it necessary to operate at weak bias fields where, as shown in Figure 30b, there is no significant field dependence of the measured gravity value.

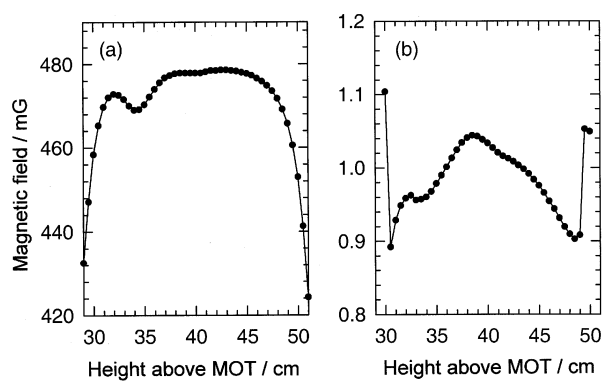

Figure 29. Magnetic field inside magnetic-shield assembly. Measured using magnetic-field-sensitive Raman transition ($1\,\mathrm{G} \equiv 10^{-4}\,\mathrm{T}$). (a) Strong bias field. Inhomogeneities dominated by solenoid field. (b) Weak bias field. Inhomogeneities dominated by other field sources.

While the data indicate that we could use bias-field values up to 50 mG, this would not be strong enough to separate the Zeeman components. To avoid partially overlapping Zeeman components we therefore choose to operate at a very small bias field, typically 2 mG, where all the Zeeman components are completely overlapped. Under these

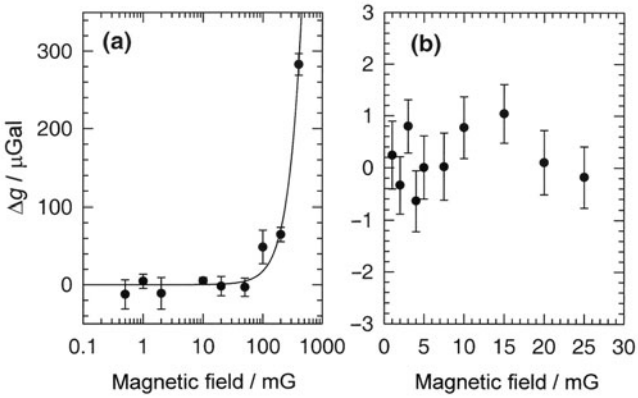

Figure 30. Magnetic-field dependence of gravity measurement ($1\,\mathrm{G} \equiv 10^{-4}\,\mathrm{T}$). The Δg axis has an arbitrary offset. The solid line in (a) is a quadratic fit.

conditions, to avoid problems with magnetic-field-sensitive transitions, only atoms in the $m_F = 0$ state can be allowed to enter the interferometer. This is taken care of by our preselection sequence (Section 3.1.4). There could be some additional problems, such as transfer into other magnetic sublevels, but these effects tend to cancel owing to the many inherent symmetries of the measurement. More importantly, they would cause shifts that depend strongly on the overall detuning Δ or the Raman beam polarizations, and no such dependence has been observed experimentally. Therefore we conclude that magnetic fields are not currently a problem.

5.1.3 Cold collisions

Atoms cooled to microkelvin temperatures have collisional cross-sections that cause large hyperfine frequency shifts [41, 42]. For ^{133}Cs atoms in a equal superposition of the $F = 3, m_F = 0$ and $F = 4$, $m_F = 0$ hyperfine ground states it has been measured by Gibble and Chu [41] to be $(-15.8 \pm 1.4)\,\mathrm{mHz}$ at a density of $(1.0 \pm 0.6) \times 10^9\,\mathrm{cm}^{-3}$.

We now estimate the effect of such cold collisions on our atom interferometer, assuming that the spatial separation of the two paths does not affect the collisions. The MOT originally captures about 5×10^8 atoms at a density of $8 \times 10^9\,\mathrm{cm}^{-3}$, but magnetic state and velocity preselection reduce the number of atoms to approximately 5×10^6 (Section 3.1.4). Because of the severe vertical velocity preselection we can neglect the spreading of the atomic cloud along the vertical axis. We furthermore neglect the initial horizontal size of the cloud and assume that its horizontal radius is proportional to the time after launch.

With these assumptions, we estimate that the average atom density is $\bar{\rho}_\mathrm{a} = 6 \times 10^7\,\mathrm{cm}^{-3}$ in the first half of the interferometer ($130\,\mathrm{ms}$ to $290\,\mathrm{ms}$ after launch) and only $\bar{\rho}_\mathrm{b} = 1.5 \times 10^7\,\mathrm{cm}^{-3}$ in the second half ($290\,\mathrm{ms}$ to $450\,\mathrm{ms}$). The resulting interferometer phase shift for our experimental parameters is

$$\Delta\phi = \kappa T\,(\bar{\rho}_\mathrm{b} - \bar{\rho}_\mathrm{a}) \approx 0.6\,\mathrm{mrad}, \tag{42}$$

where κ is the frequency shift cross-section. This corresponds to a gravity offset of 0.2 μGal.

The effect would be greater by approximately a factor of 15 without vertical velocity preselection, and greater by another factor of 3 without magnetic sublevel preselection (the cross-sections for the other m_F states are somewhat smaller).

5.2 Raman beams

5.2.1 Determination of the caesium D_2 *wavelength*

From the formula $\Delta\phi = k_\mathrm{eff} g T^2$ it is apparent that to obtain a gravity value one has to know the Raman wave vector $k_\mathrm{eff} = |\boldsymbol{k}_2| + |\boldsymbol{k}_1|$ or, in other words, the $6S_{1/2}$, $F = 3$ to $6P_{3/2}$, $F' = 4$ transition frequency to the same level of accuracy. At the time of our measurement this frequency had only been determined by several wavemeter measurements with a range of uncertainties $\sim 8\,\mathrm{MHz}$ (Figure 31).

Using the frequency value by Avila et al. [44] resulted in a large discrepancy between the atom interferometer gravity value and the one measured by the falling corner-cube gravimeter (see Section 4.4). However, Hänsch's group at the Max Planck Institute for Quantum Optics has recently performed a direct

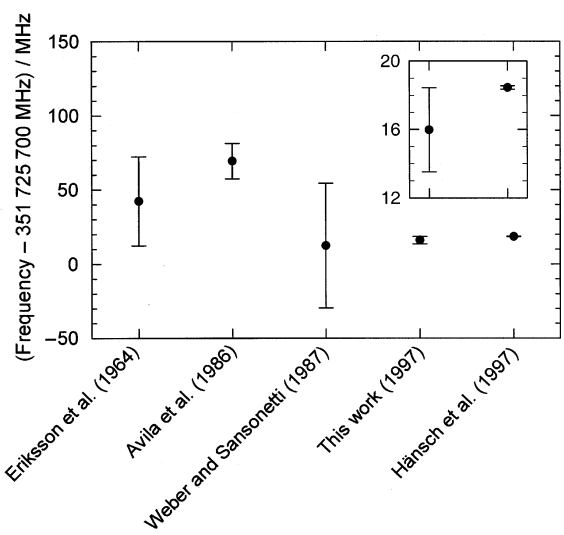

Figure 31. Measured values for the caesium D_2 frequency ($6S_{1/2}$, $F = 3$ to $6P_{3/2}$, $F = 4$ transition). Eriksson et al. [43] and Avila et al. [44] used wavemeters to measure the D_2 wavelength. Weber and Sansonetti [45] performed a wavemeter measurement of the D_1 wavelength; the large error bars shown are due to the transfer. Hänsch et al. measured the D_2 frequency directly. Our own value is calculated by comparing atom interferometer and falling corner-cube measurements of g. Inset shows a magnified view of the last two points.

measurement of the caesium D_2 frequency and the preliminary result is about $50\,\mathrm{MHz}$ lower than the weighted average of the wavemeter measurements. We have since then performed another 48-hour monitoring of g with the improvements outlined in other parts of this section. Using the recently measured value of g and the value of the D_2 frequency measured by Hänsch's group, the discrepancy between the atom interferometer and classical gravity measurements is found to be (7 ± 7) μGal (see Section 6), assuming that g (in our laboratory at Stanford) did not change in the six months between the measurements (Figure 31).

5.2.2 Vertical alignment

To obtain the correct absolute gravity value the Raman wave vector has to be exactly parallel to the local vector of gravity. The relation

$$\Delta\phi = \boldsymbol{g} \cdot \boldsymbol{k}_{\mathrm{eff}} T^2 = |\boldsymbol{g}||\boldsymbol{k}_{\mathrm{eff}}| T^2 \cos\theta \tag{43}$$

shows that misalignment by an angle θ will result in too low a gravity value. For small angles this error is quadratic in θ. In order to achieve an accuracy of 1 μGal, the deviation of the Raman beams from vertical must be less than 45 μrad. This direction is effectively determined by the orientation of the retroreflection mirror, which in turn is controlled by an active tilt-stabilization system (Section 3.3.3). The electronic bubble level is mounted rigidly to the mirror and has been calibrated to approximately 5 μrad. The tilt error should therefore remain well below the critical value.

It is possible that the calibration of the bubble level could change over time. Furthermore, a wedge in the top vacuum window could deflect the wave vector at the location of the atoms relative to its direction at the mirror surface. To avoid this, we use windows that are plane parallel to better than 10 μrad, but they might be deformed by the pressure difference.

We tested the vertical alignment by tilting the whole experimental set-up over a substantial angular range. Using (43) to fit the resulting change in g we determined the correct vertical orientation. As the result agrees to 10 μrad with the calibration of the tilt sensor, we expect the resulting gravity error to be less than 0.05 μGal.

5.2.3 Retroreflection alignment

Any failure to properly retroreflect the Raman beams affects the measurement result by changing the magnitude of the wave vector. If the angle of incidence of the incoming Raman beams at the mirror deviates from 90° by an angle ϑ we have

$$|\boldsymbol{k}_{\mathrm{eff}}| = (|\boldsymbol{k}_1| + |\boldsymbol{k}_2|) \cos\vartheta, \tag{44}$$

which for small angles causes an error quadratic in ϑ.

Currently, this retroreflection is checked rather crudely by aperturing the Raman beams down to $2\,\mathrm{mm}$ at a distance of approximately $3\,\mathrm{m}$ from the retroreflection mirror and then centring the reflected beams on the same aperture. This can be done quite easily with an uncertainty of less than $0.2\,\mathrm{mm}$, corresponding to an angle of 30 μrad and a measurement uncertainty of less than 1 μGal.

To validate this alignment procedure we intentionally tilted the incoming Raman beam. Using (44) to fit the observed change g, we find a tilt angle for proper retroreflection that differs from the one obtained by the aperture method typically by less than 40 μrad.

Even after using the aperture alignment method we must still expect an uncertainty as large as –1 μGal. As the angular direction of the incoming Raman beams tends to drift, this alignment has to be checked frequently.

5.2.4 Wavefront curvature

The theoretical derivation of the interferometer phase shift assumes the Raman beams to be ideal plane waves. Our Raman beams, on the other hand, are Gaussian laser beams (Bessel beams when a pinhole is used for spatial filtering). The wavefronts of Gaussian beams are spherical and can be approximated by plane waves only for distances from the beam waist that are much smaller than the Rayleigh length $z_0 = \pi w_0^2/\lambda$. For our $2.5\,\mathrm{cm}$ diameter Raman beams the Rayleigh length is $575\,\mathrm{m}$.

We have investigated this effect by moving the output lens of the beam-expansion telescope ($1.5\,\mathrm{m}$ focal length) over some distance, which causes substantial changes of the wavefront curvature in the interaction region. The parabolic dependence of the observed shift agrees with theoretical predictions [16]. Our standard collimation procedure using a shear-plate collimation tester should therefore ensure that errors arising from wavefront curvature are well below 1 μGal.

5.2.5 Speckle

We have also considered the effect of the more local wavefront perturbations caused by optical scatterers and imperfect optical elements. They can be very complex and in more severe cases are generally referred to as speckle [46, 47].

We have investigated this effect by looking for observable changes while intentionally making things worse – here by sprinkling baby powder on to a glass plate inserted into the beam path. Figure 32 shows the effect of this rather drastic procedure as observed by a CCD chip illuminated directly by the Raman beams.

Amazingly, this procedure did not result in any significant gravity offsets (Figure 33), even in extreme cases with enough powder on the window to reduce the interferometer contrast to less than $25\,\%$. Therefore, given that our optics are much cleaner, we rule out speckle as the cause of any significant error.

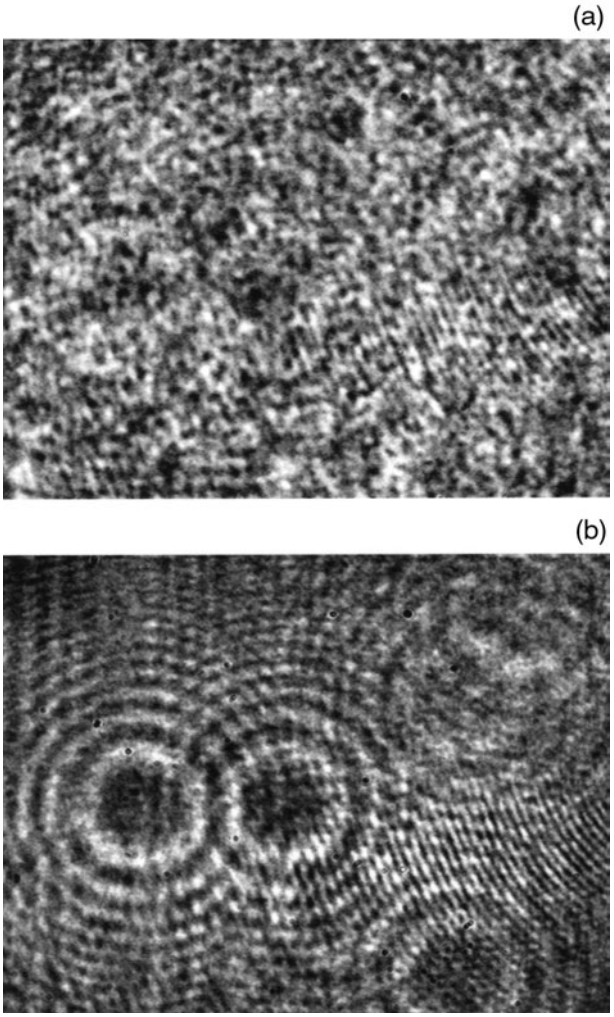

Figure 32. Local intensity variations of the Raman beams, observed by direct illumination of a CCD (1/2-inch wide) inserted into the beams after they passed through the vacuum chamber. (a) The speckle pattern generated by a glass plate covered in baby powder inserted into the beams which reduces the interferometer fringe contrast to 24 %. (b) The intensity distribution without this additional plate. Only a few diffraction rings, caused by small particles of dirt on the lower vacuum chamber window, are visible.

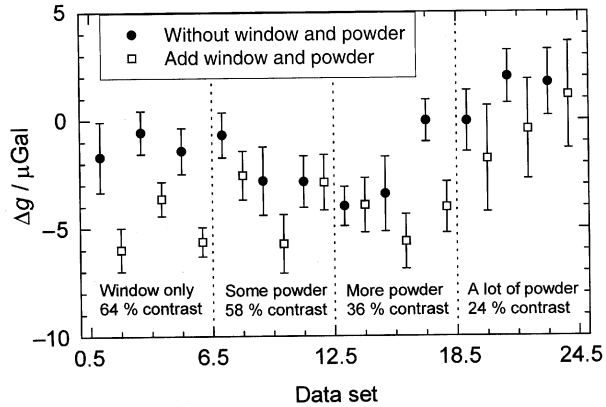

Figure 33. Effect of speckle on the measured gravity value. The open squares are measurements taken with a glass plate, covered with variable amounts of baby powder, inserted into the Raman beams before the vacuum chamber. The solid dots are measurements without this additional plate. They are necessary for comparison, as the measured gravity values are not corrected for environmental changes and ocean-loading effects. While adding more powder substantially decreases the fringe contrast, it has no significant effect on the measured gravity value. The small offset between the two data sets is caused by the small wedge of the glass plate.

5.2.6 Refractive index of background vapour

The refractive index of the background vapour can potentially modify the wave vector of the Raman beams and thereby cause a shift in the measured gravity value.

Generally one would assume the refractive index inside an ultra-high-vacuum system to be sufficiently close to unity. However, a substantial fraction of the vapour pressure is due to caesium vapour. This will cause a substantially increased refractive index for the Raman beams as their frequency is close to a caesium transition. We therefore estimate at what partial pressure of caesium this would become a problem.

The refractive index is related to the complex absorption coefficient by $\Delta n_{\mathrm{ref}} = n_{\mathrm{ref}} - 1 = \mathrm{Im}\,\alpha/k$ [48]. We have

$$\mathrm{Im}\,\alpha = \frac{n\sigma}{2}\,\frac{1}{(1+s)}\,\frac{\delta}{(\Gamma/2)}\,\frac{1}{1+(\delta/[(\Gamma/2)\sqrt{1+s}\,])^2}, \tag{45}$$

which in the far detuning limit reduces to

$$\mathrm{Im}\,\alpha = \frac{n\sigma}{2}\,\frac{(\Gamma/2)}{\delta}. \tag{46}$$

Here, s is the saturation parameter, n the atomic number density and σ the absorption cross-section on resonance, and δ is the detuning. For circularly polarized light, and after averaging over all excited states, the absorption cross-section is $2.37 \times 10^{-9}\,\mathrm{cm}^2$ for atoms equally distributed among all magnetic sublevels of the $F = 4$ state and $2.65 \times 10^{-9}\,\mathrm{cm}^2$ in the case of the $F = 3$ state. We approximate both numbers by $2.5 \times 10^{-9}\,\mathrm{cm}^2$, assume that each Raman beam interacts only with atoms in one of the two hyperfine states and that the background atoms are equally distributed among these. We also neglect the Doppler broadening as well as the excited-state hyperfine splitting, use the relation $10^{-9}\,\mathrm{Torr} \sim 3.2 \times 10^7\,\mathrm{atoms/cm}^3$ for caesium at room temperature, and assume a detuning Δ of 1 GHz. This leads to the relation $(\Delta g/g)/p = 0.8\,\mathrm{Torr}^{-1}$ and implies that the partial pressure of caesium has to be less than $1.2 \times 10^{-9}\,\mathrm{Torr}$ to keep the gravity offset below 1 μGal.

Our system, which is a hybrid between vapour-cell MOT and beam-loaded MOT, exhibits large differences in the local caesium density. It is therefore difficult

to estimate the partial pressure of caesium in the interaction region itself, but absorption measurements indicate that it is less than 10^{-10} Torr (10^{-8} Pa). The refractive index of the background vapour should therefore cause a shift of less than 0.1 μGal.

5.3 Frequency control and pulse timing

Among all physical quantities, time and frequency are generally the most accessible to current experimental methods and measurable to the highest precision and accuracy. Many high-precision experiments therefore transform the required measurement into one of time and frequency. Following this philosophy, our atom interferometer effectively measures g by converting it into a phase-shift value that depends only on the frequency of the Raman beams and the time between light pulses.

Certain features of our experiment further tailor this process to operate in a regime where current technology works best. The use of stimulated Raman transitions alleviates the requirement of millihertz-level optical-frequency control and requires such precision only in the microwave and radio-frequency range where it is much easier to achieve. Changing the Raman frequency difference to nearly cancel the gravity-induced Doppler shift is necessary to keep the atoms in resonance, but it also reduces the sensitivity to the timing of the light pulses. This is highly beneficial, since frequencies can be controlled even better than short time intervals.

Even with these methods, timing and frequency control are not trivial at the level required to achieve an accuracy of 1 μGal. In marked contrast to most atomic physics experiments where rf and microwave components can usually be considered ideal, we therefore have to scrutinize every single component of the frequency control and timing system.

5.3.1 Frequency-dependent phase shifts

Changing the Raman frequency to keep the atoms in resonance can introduce systematic errors arising from non-linear, frequency-dependent phase shifts (group delay). Such phase shifts can be introduced through filters, spurious frequency components from the direct digital synthesis process, interference effects from back reflections of rf power or rf leakage along alternate paths.

We attempted an independent measurement of this effect. Ideally, such a measurement would look at the system as a whole, starting with the mathematical description of the frequency chirp and then analysing the phase shift of the Raman beams themselves. For technical reasons, however, mainly stemming from the nature of the DDS process, measurements have to be divided into at least two parts.

To determine the phase shifts of the Tektronics DDS synthesizer we use a fast digital scope to measure the delay from the SYNC signal indicating the beginning of the rf pulse (starting at zero phase) to the next zero crossing of the output waveform. Since the SYNC signal is internally synchronized with the synthesizer's phase accumulator, and therefore the *synthesized* phase, this measurement of the *actual* output phase allows us to determine the frequency-dependent phase shift introduced by the amplifiers and filters internal to the synthesizer. The absolute length of the delay is not important, only the change as a function of frequency. Any additional, fixed delay would only appear as a phase shift that changes linearly as a function of frequency and is not relevant for our experiment. Figure 34a shows the measured phase shifts. For our standard chirp centred around 40 MHz, we find an additional interferometer phase shift $\Delta\phi_{\mathrm{rf}} = \phi(43.68\,\mathrm{MHz}) - 2\phi(40\,\mathrm{MHz}) + \phi(36.32\,\mathrm{MHz}) \simeq 15\,\mathrm{mrad}$, corresponding to 4 μGal gravity offset. For some of our most recent measurements we have therefore moved the chirp centre frequency to 16 MHz where the non-linear phase shift is minimal and we expect a gravity offset of only 0.4 μGal.

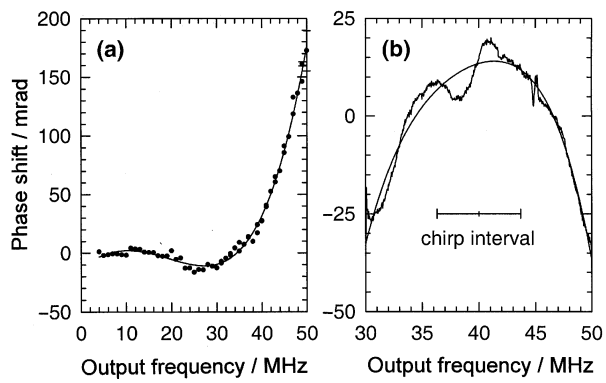

Figure 34. Frequency-dependent phase shifts. (a) Tektronics AFG 2020 DDS synthesizer with internal 100 MHz anti-aliasing filter. Measured using fast digital scope. (b) Remaining system, from output of the DDS synthesizer to Raman beams. Measured using vector network analysis. In both cases the offset as well as the linear term are arbitrary and not relevant for the experiment.

The remaining system (starting at the output of the DDS synthesizer) can be investigated by standard vector network analysis. The result is plotted in Figure 34b. The global curvature of this response graph would correspond to a gravity offset of 2.2 μGal. The smaller scale wiggles visible in the graph would cause additional microgal scale offsets depending on details such as the central frequency of the chirp or the time between Raman pulses. However, as we had to introduce many additional components to perform this measurement we cannot be sure (without much more involved calibration procedures) if all of the observed features are due to components used in the experiment.

As the measurements described above are not fully conclusive, we use a special property of the atom

interferometer to measure and cancel the effect of non-linear phase shifts. Reversing the direction of the photon recoil also means changing the direction of the rf chirp and the sign of the gravity offset. One can therefore alternate between the two interferometer configurations and calculate the average value, which should be free of phase-shift related offsets. The differential value is actually a measurement of the non-linear phase shift. Figure 35 illustrates the effectiveness of this method. However, changing the recoil also introduces some other changes, such as slightly altered trajectories or pulse times, which may cause additional offsets. In order to reverse the chirp exactly, with the same centre frequency, it is for example necessary to apply the middle interferometer pulse *after*, instead of *before*, the atoms reach the apex of their trajectory (or vice versa). This is because we use retroreflected Raman beams and if this pulse were to be applied exactly at the apex, it would also drive unwanted Doppler-insensitive Raman transitions. To account for this, we often take gravity data for all four possible combinations of recoil direction and pulse time, as shown in Figure 36.

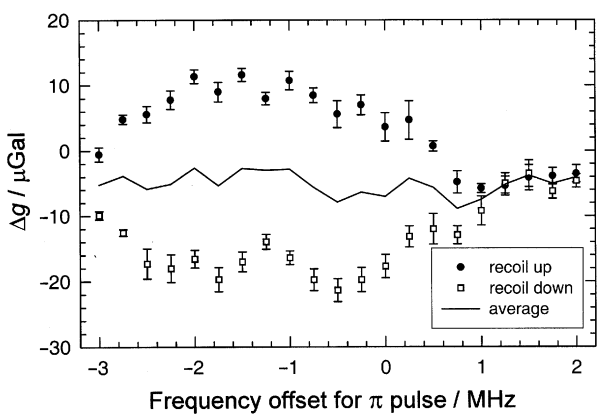

Figure 35. Effect of frequency-dependent phase shifts. The measured gravity value changes if the centre frequency of the DDS synthesizer chirp is changed from its usual $40\,\mathrm{MHz}$ value (compensated for by changing the offset frequency of the Raman laser phase-lock, see Section 3.2.2). Reversing the direction of the photon recoil also means reversing the direction of the chirp and the sign of the gravity shift. The average value is therefore much less sensitive to this effect. Interferometer pulse separation $T = 160\,\mathrm{ms}$.

In summary, the method of alternating between differential recoil configurations can substantially reduce the sensitivity to imperfections of the rf system, but the remaining uncertainties are still some of the most important to affect the atomic fountain gravimeter. From the uncertainty of measurements using different centre frequencies and recoil directions we estimate a residual uncertainty of 2 μGal.

5.3.2 Timing of Raman pulses

From the formula $\Delta\phi = gk_{\mathrm{eff}}T^2$ one would naively expect that a 1 part in 10^9 measurement of g would

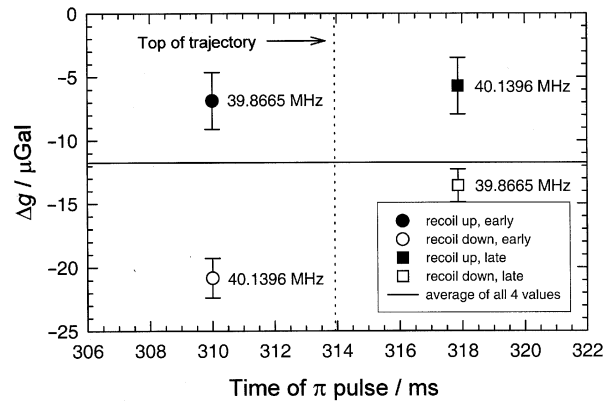

Figure 36. Changing the direction of photon recoil and rf chirp cancels most rf phase-shift-related gravity offsets. But one has to change either the time of the interferometer π pulse (early = before apex of trajectory, late = after) or the centre frequency of the rf chirp (see graph). The graph shows measurement results for all four possible combinations. They are usually averaged to cancel most of the effect.

require knowledge of the interferometer pulse spacing T with similar accuracy. For our typical $160\,\mathrm{ms}$ spacing this corresponds to only $80\,\mathrm{ps}$, a requirement that is essentially impossible to fulfil using standard beam-switching methods. Fortunately, changing the Raman frequency to cancel most of the gravity-induced Doppler shift makes the timing requirements much less severe.

Assume that we determine g by finding the chirp rate at which a continuous chirp of the Raman frequency *exactly* cancels the gravity-induced Doppler shift ($\Delta\phi = 0$). This condition is obviously independent of the interferometer pulse spacing T. Changing T therefore does not affect the position of the central interferometer fringe, which indicates when this condition is fulfilled. It only changes the fringe period.

The situation is slightly different when the continuous chirp is approximated by switching between three fixed frequencies (Section 3.2.2). In that case we are once again sensitive to the pulse timing, but now small timing errors enter only quadratically. For small errors δt_i in the time of the three interferometer pulses we find

$$\frac{\Delta g}{g} = \frac{\delta t_1^2 + 2\delta t_2^2 + \delta t_3^2}{2T^2}, \tag{47}$$

which can be shown by first choosing a frequency step Δ that exactly cancels the acceleration-induced chirp (see Figure 13) for a fixed-pulse separation T and then calculating the effect of small timing perturbations. This result is also confirmed by experimental data [23]. As we can easily achieve a timing accuracy of $100\,\mathrm{ns}$, the possible error due to pulse timing is less than 0.001 μGal.

5.3.3 Length and intensity of Raman pulses

Variations in the length and intensity of the Raman pulse are other potential sources of trouble. For a single

pulse the effect of such variations has been worked out in [16]. A priori, it is large enough to be of concern if the Raman beam intensities, as in our set-up, are not actively stabilized. But as long as these variations occur for all three interferometer pulses simultaneously, the effects cancel almost completely due to the symmetry of the $\pi/2 - \pi - \pi/2$ pulse sequence. Even when we use a $\pi/4 - \pi/2 - \pi/4$ pulse sequence, corresponding to a rather drastic error in intensity, the measured value of g only shifts by a few microgals.

Other tests confirm that the measured gravity value remains unchanged when short, high-intensity pulses are replaced with longer, low-intensity pulses while keeping the pulse area constant and maintaining the correct time between pulses. Intensity variations among the pulses of the interferometer are sufficiently small to be neglected. We therefore conclude that the uncertainty arising from these effects is less than 1 μGal.

5.3.4 Chirping and detuning of Raman pulses

We often perform gravity measurements by simply switching between fixed Raman frequencies instead of applying a continuous chirp. In this case the atoms, in their own free-falling reference frame, see an apparent chirp of the Raman beams during each single pulse. For finite-length pulses this will cause phase shifts [23]. For typical experimental parameters the effect of these shifts is rather small as a result of the symmetry of the $\pi/2 - \pi - \pi/2$ pulse sequence.

It is still preferable to eliminate this concern completely by using the second phase-locked loop in our set-up (Section 3.2.2) to cancel the gravitational chirp during each individual pulse. The same method can also be used to introduce deliberately chirps of the wrong sign or magnitude. The result is shown in Figure 37 and clearly confirms the theoretical prediction that the measurement is rather insensitive to such chirps.

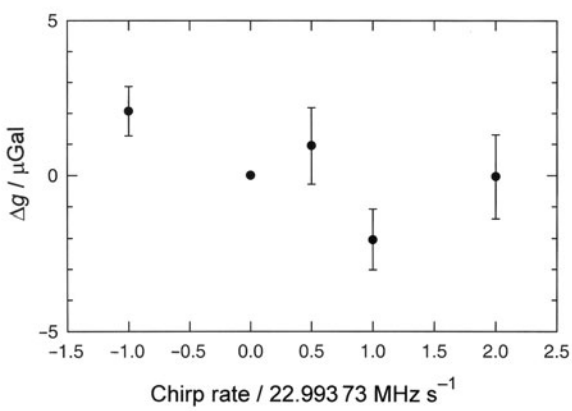

Figure 37. Effect of chirp during Raman pulses. We can introduce chirps of the Raman frequencies during each interferometer pulse. A chirp rate of $22.993\,73\,\mathrm{MHz/s}$ exactly cancels the gravitational chirp, the ideal situation. But even for very different chirp rates the gravity value is almost unaffected.

Using the $\pi/2 - \pi - \pi/2$ pulse sequence makes the interferometer intrinsically insensitive to detunings from the exact Raman resonance condition, a feature absolutely necessary considering the velocity spread of the atoms and the corresponding Doppler shifts. This does not rule out small residual offsets when the Raman frequency is not properly centred with respect to the atomic Doppler distribution. We found experimentally that this gravity offset is linearly proportional to the average detuning with a slope of ∼1 μGal/kHz.

We now estimate if frequency offsets of this size could occur. As all the Raman frequencies are referenced to a Loran-C receiver, and thereby ultimately to an atomic clock, they are generally known to better than 5×10^{-12}, or $0.05\,\mathrm{Hz}$. The resonance can easily be located to better than $100\,\mathrm{Hz}$ and, because of the velocity preselection, is largely insensitive to small changes in the atomic velocity distribution. This resulting gravity offset would be approximately 0.1 μGal.

Bordé [49] predicted that the measured g in our experiment should depend on the detuning of the laser from the atomic resonance *during* the pulse, similar to the corrections of a few parts in 10^2 using dynamic diffraction treatment in neutron interferometer experiments [50]. However, the above results show that the effects are at most at a relative level of 10^{-9} in our experiments. This is probably due to the difference in scale between the two experiments and also the symmetric nature of our interferometer.

5.3.5 Changes in the Raman wave vector

When we change the Raman frequency $\omega_{\mathrm{eff}} = \omega_1 - \omega_2$ to compensate for the gravity-induced Doppler shift, we can either change ω_1, ω_2, or both. If the two frequencies are not changed by opposite amounts, then the resulting change in the effective Raman wave vector k_{eff} will affect the interferometer measurement in several ways. Most importantly, it will affect the phases read in during the interaction of the atoms with the light field. Furthermore, it will modify the atomic trajectories, since different amounts of momentum are transferred during beam-splitter and mirror pulses.

For our experiment the changes in the atomic trajectories cause only a minute shift of $\sim 10^{-18}\,g$ and can be neglected. The changes in the Raman phases, however, can be important.

Assume that we chirp the frequencies of the two Raman beams with chirp rates α_1 and α_2, such that the chirp rate $\alpha_1 - \alpha_2$ of the Raman frequency exactly cancels the gravity-induced Doppler shift. Then the gravity value calculated from the condition $k_{\mathrm{eff}} g = (\alpha_1 - \alpha_2)$, where k_{eff} is the Raman wave vector at the moment of the π pulse, has to be corrected by

$$\frac{\Delta g}{g} = \frac{2v_0}{c}\,\frac{\alpha_1 + \alpha_2}{\alpha_1 - \alpha_2}, \tag{48}$$

where v_0 is the atomic velocity at the time of the π pulse.

The correction is zero if both Raman beams are either chirped by equal amounts in opposite directions, or if the central Raman pulse is applied exactly at the top of the fountain. However, for our typical fountain configuration we chirp only one of the Raman beams and we have to make the interferometer slightly asymmetric in order to prevent the central Raman pulse from driving Doppler-insensitive transitions. The velocity v_0 corresponding to this asymmetry is about $0.04\,\mathrm{m/s}$ and results in a 0.3 μGal correction. If, instead of using an almost symmetric fountain, we simply dropped the atoms (using the same pulse spacing), then this effect would cause a much larger offset, of 10.4 μGal. This effect has been experimentally verified.

Equation (48) can also be derived by including retardation effects into the analysis of the interferometer. The effect described here is therefore related to the finite speed of light correction necessary for falling corner-cube gravimeters [51].

5.4 Synchronous noise

In contrast to a continuous measurement, noise synchronized with the data-acquisition process of a pulsed experiment does not generally average out over time but causes a systematic offset. Vibrations, stray light and electromagnetic interference are just a few examples of the multitude of possible culprits. Modifying the timing of the experiment to ascertain which of them are responsible is limited by the existence of many timing constraints and the unavoidable interdependence of timing parameters with others, for example spatial position and frequency.

5.4.1 60 Hz *line noise*

If the experiment is line-triggered, then line noise (60 Hz) is an obvious source of trouble. We observed a ±5 μGal sinusoidal oscillation of our gravity value as a function of the delay between interferometer pulse sequence and line trigger. We therefore do *not* line trigger our experiment during gravity measurements in order to make the effect average out over time, assuming that the system's response to the 60 Hz disturbance is linear. In this mode of operation the effect then becomes just another source of noise.

5.4.2 Magnetic sensitivity of seismometer

The central component of the active vibration-isolation system is a commercial, low-frequency accelerometer, not specifically designed for use in feedback systems but for applications in seismology and geophysics. When performing measurements for these purposes, the seismometer is usually located either in the field or in specially constructed seismic vaults. In neither case is it subjected to substantial electromagnetic interference and accordingly the design of the instrument does not include special precautions to control such influences.

The situation in the laboratory is very different, with the seismometer exposed to external disturbances of various types. Many of them are synchronized with the experimental cycle and could therefore lead not just to increased noise but also to systematic errors. In the following we analyse the effect of pulsed magnetic fields, to which the seismometer is particularly susceptible.

The seismometer will mistakenly interpret any change in the external magnetic field as an acceleration and the feedback loop, attempting to compensate, will introduce a real acceleration of the retroreflection mirror in the opposite direction. Since the acceleration of the atoms is measured relative to this mirror, this in turn will cause a systematic measurement error. The precise size of this error will depend on details such as the parameters of the feedback loop or the duty cycle of the magnetic field, but it will generally be of the same magnitude as the fake acceleration.

For our set-up, pulsing of both the trapping coils and the Helmholtz coil assembly can result in field strengths of several hundred milligauss at the location of the vibration isolator. Together with the measured magnetic-field sensitivity of ∼150 μGal/G of our particular version of the CMG-3 seismometer, this implies unacceptably large errors. The seismometer was therefore fitted with a magnetic shield (Section 3.3.1) even before assembling the system, reducing its magnetic-field sensitivity by at least a factor of 10. However, the magnetic properties of this shield resulted in additional magnetic forces on the suspended mass. To test the effect, we extended the normal cycle time of the experiment from $1.3\,\mathrm{s}$ to $2.0\,\mathrm{s}$ by inserting a delay before reloading the MOT. During this time interval we can pulse on the Helmholtz coils to produce a magnetic field either equivalent to that used for $\sim 600\,\mathrm{ms}$ during the normal experimental cycle, or with opposite sign.

Graph (a) in Figure 38 shows that even with a magnetic shield around the seismometer itself the measured gravity value is strongly affected by these pulses. If we pulse on the MOT trapping coils during the additional time interval, we find a similar effect. For a standard measurement sequence, the expected gravity shift due to this effect should be of approximately the same magnitude. To reduce this effect to acceptable levels we have enclosed the whole vibration-isolator assembly with an additional magnetic shield. Graph (b) indicates that this reduces the effect to below our current measurement resolution.

5.5 Fundamental

5.5.1 Coriolis force and Sagnac effect

Atom interferometers are intrinsically sensitive to all types of inertial force. Our set-up is specifically

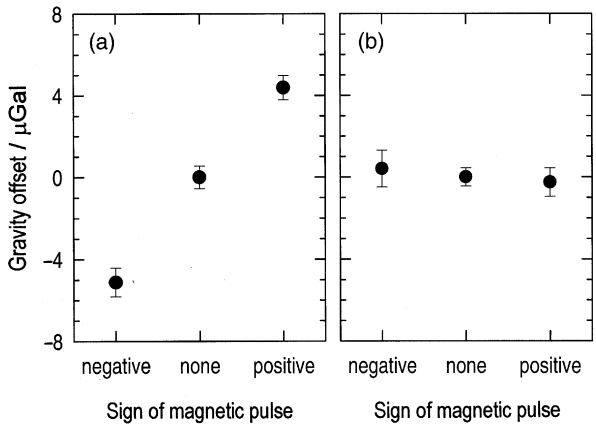

Figure 38. Effect of pulsed magnetic fields arising from the magnetic-field sensitivity of the seismometer in the vibration-isolation system (a) before and (b) after adding an additional magnetic shield.

designed to be sensitive only to the acceleration due to gravity, but even small misalignments can make it susceptible to Coriolis forces. Some Coriolis forces, for example those caused by rotational noise of the experimental set-up, are avoidable. But as our experiment is performed in a terrestrial laboratory, the Coriolis forces due to the Earth's rotation are always present and can cause systematic errors if not properly accounted for.

The atoms in the atomic fountain, though laser-cooled, still have a certain horizontal velocity spread. If we trace the trajectory of an atom that has a non-zero initial horizontal velocity, we find that the two interferometer paths enclose a finite *spatial* area (Figure 39). Whenever this happens the atom interferometers become sensitive to rotations. This is also known as the Sagnac effect. For a Mach-Zehnder type of configuration the resulting phase shift is given by

$$\Delta\phi = \frac{2m}{\hbar}\boldsymbol{\Omega}\cdot\boldsymbol{A} = 2\boldsymbol{\Omega}\cdot(\boldsymbol{v}_0 \times \boldsymbol{k}_{\mathrm{eff}})T^2, \tag{49}$$

where $\boldsymbol{v}_0$ is the initial velocity and $\boldsymbol{A}$ a vector normal to the enclosed area and with a length proportional to its size [19]. The corresponding gravity offset,

$$\Delta g = 2\boldsymbol{\Omega}\cdot(\boldsymbol{v}_0 \times \hat{\boldsymbol{v}}_{\mathrm{rec}}), \tag{50}$$

is exactly what one would find by calculating the classical Coriolis acceleration. All these formulas also remain valid in the presence of a constant gravitational field.

Substituting the Earth's rotation rate $\Omega = 7.29 \times 10^{-5}\,\mathrm{rad/s}$ (one rotation per one 23 h 56 min 4 s sidereal day, not one solar day) and the geographical latitude of our laboratory (37° N.) we find that the initial east-west transverse velocity has to be less than $8.6 \times 10^{-5}\,\mathrm{m/s}$ to achieve 1 μGal accuracy. In other words, since the total time of flight is $610\,\mathrm{ms}$, the atoms have to return to within 52 μm of their initial position.

The atoms in the atomic fountain have a certain velocity distribution, so what matters is the *average* horizontal velocity of all the *detected* atoms. Within limits, we can adjust this average velocity by changing the size and position of the detection region, either by moving the excitation beam or the aperture in front of the PMT. Figure 39 shows that we can easily detect the Earth's rotation using this method. It also demonstrates that the effect is strong enough to cause large systematic errors even for small misalignments.

It would be possible to reduce this effect by transverse velocity preselection, but we could not use

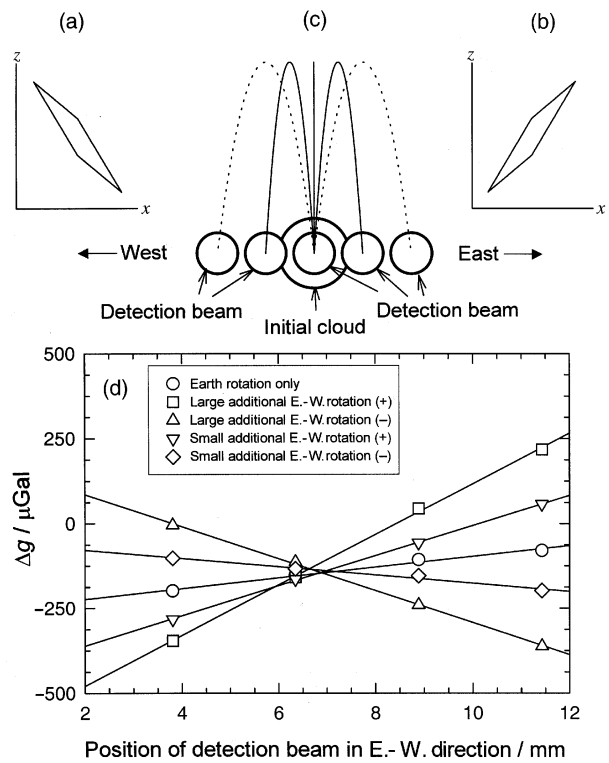

Figure 39. Effect of Coriolis forces on the measured gravity value. (a) and (b) For atoms with non-zero horizontal velocities the two interferometer paths enclose a *spatial* area, which makes the interferometer sensitive to Coriolis forces. (c) By moving the detection beam, we can preferentially select atoms with different horizontal velocity components, resulting in a different magnitude and sign of the interferometer phase shift (which corresponds to a certain gravity offset) caused by the same rotation. (d) Using this method, we can easily detect the Earth's rotation (circles). The slope is about 150 μGal/cm instead of the expected 196 μGal/cm. This difference can be quantitatively explained by the aperturing effect on the Gaussian distribution of the atom cloud. The other symbols show the additional gravity shifts produced by intentionally rotating the optical table clockwise and counter-clockwise. The intersection of the lines marks the detection-beam position at which the Coriolis effect vanishes. All measurements use a pulse separation of $T = 160\,\mathrm{ms}$.

this method owing to lack of optical access. Instead, we use the interferometer signal itself to achieve the proper alignment. We can find the correct position of the detection region – where the Coriolis effect vanishes – by introducing additional, well-defined rotations. Figure 39 demonstrates how this can be done by rocking the table so that it rotates clockwise and counter-clockwise during alternating launches of the atoms.

This method can in principle provide the required alignment accuracy, provided that the experimental set-up is rotated around a point directly below the sensitive spot of the vibration isolator. Otherwise even the performance of the 1-D vibration-isolation system is not sufficient to suppress the vertical accelerations introduced by the shaking and we find an erroneous value for the Coriolis-insensitive detection-beam position.

It is also important to correctly adjust the phase of the sinusoidal table rotation with respect to the interferometer π pulse. Otherwise, if $\dot{\Omega} \neq 0$ at the time of the π pulse, the measurement becomes sensitive to horizontal displacements of the atoms with respect to the sensitive spot of the vibration isolator (Section 4.2.6). This can in turn result in a wrong value for the correct detection-beam position.

Finally, since our laboratory is not *exactly* north-south oriented, it is also important to perform the Coriolis calibration by shaking the table along both axes.

We have found that the direction of the launch changes over time scales of several hours. Long-term gravity measurements using the current set-up therefore require frequent checks of the Coriolis calibration. Overall, sensitivity to Coriolis forces remains the weak point of our current apparatus, introducing an uncertainty of $\sim$2 μGal. This problem would be greatly reduced if the entire apparatus could be rotated about the vertical axis.

5.5.2 Gravity gradients

Gravitational fields are typically not uniform but exhibit some degree of spatial variation. The inverse-square-law dependence of the Earth's gravitational field results in a vertical gravity gradient of approximately 300 μGal/m at its surface. Variations in the local mass distribution due to geological features can change this value significantly.

This effect has to be taken into account when analysing the interferometer data as along their trajectory the atoms encounter changes in gravity which are large compared with the targeted measurement accuracy of 1 μGal. As long as the gravity gradient can be considered linear over the $\sim 10\,\mathrm{cm}$ length scale of the atomic fountain, we easily treat it using (14) from the theory section. Non-linearities in the gravity gradient, on the other hand, would substantially complicate the analysis of data from the atom interferometer. These non-linearities could be caused by local mass distributions in the building, the laboratory or even the measurement apparatus itself.

We have measured this gradient experimentally, using the atom interferometer. To vary the vertical position of the interferometer within the measurement region, we launch the atoms to different heights by changing the launch velocity. We also have to change the time of the launch, which allows us to keep all other parameters (Raman frequency, pulse timing, ...) constant in order to ensure that any small systematic effects associated with them cancel. The particular method that we employ to achieve this rules out using velocity preselection, the main consequence of which is a loss of fringe contrast.

We also rotate the optical table to cancel the Earth's rotation. Otherwise, changes in the atomic trajectories could make the measurement sensitive to Coriolis forces. Since the magnetically shielded region has a finite length, we have to reduce the time T between Raman pulses to $90\,\mathrm{ms}$ in order to obtain a smaller fountain and gain some space to move it up and down. This substantially reduces the measurement sensitivity. Together with the fact that the table rotates in the correct direction only for every second launch, this leads to measurement times longer by a factor of 20 than under normal circumstances and makes it very difficult to obtain a good measurement resolution.

Figure 40 shows the result of the measurement, demonstrating that the gravity gradient in the measurement region is essentially linear. From a linear fit of the data we obtain a value of (278 ± 31) μGal/m. This agrees well with the independently measured (293 ± 10) μGal/m outside the vacuum chamber (see below). Given these results, we conclude that the gravity gradient inside the measurement region is well enough known not to cause problems at our current measurement accuracy.

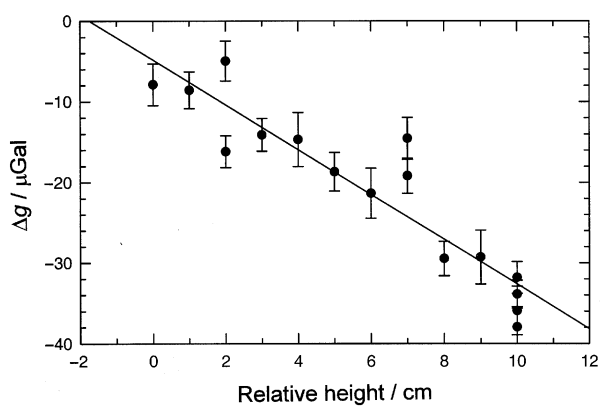

Figure 40. Measurement of gravitational gradient inside the interaction region. The data are obtained using a smaller atomic fountain (pulse spacing $T = 90\,\mathrm{ms}$) and moving it up and down inside the measurement region by adjusting the launch parameters. The solid line is a linear fit.

Future gradient measurements could use different interferometer configurations which are insensitive to a constant gravitational field but for which the interferometer phase shift is a direct measure of the gravity gradient. Two possible configurations are the "double-diamond" geometry [52] and the "triple-diamond" arrangement [53]. The latter has the advantage of being totally insensitive to Coriolis forces and makes better use of the time the atom spends in the finite-size measurement region. However, these interferometers lack the symmetry of the standard Mach-Zehnder configuration and the resulting high degree of common-mode rejection for many systematic effects. Using them therefore requires a careful re-evaluation of these effects.

Gravity gradients are also important in the comparison of gravity values obtained by different instruments. Since two instruments cannot (at least for simultaneous measurements) operate at exactly the same location, one has to transfer both gravity values to a common spatial reference point.

Since we have substantial masses in our laboratory, for example the optical table, we could not assume the gravity gradient to be linear and constant. The atomic fountain apparatus also could not be moved, precluding access to the exact measurement site. We therefore moved a LaCoste-Romberg type G gravimeter between the reference location of the FG5 absolute gravimeter measurement ($\sim 1.308\,\mathrm{m}$ above floor level, $\sim 2.5\,\mathrm{m}$ horizontally from the atomic fountain) and a point at the same height as the top of the atomic fountain and as close to it as possible ($1.810\,\mathrm{m}$ above floor level, $\sim 30\,\mathrm{cm}$ horizontally from the atomic fountain). Repeating this procedure fifteen times, we measured a gravity difference of (147 ± 5) μGal, corresponding to a linear gravity gradient of (293 ± 10) μGal/m. The relatively large error mainly arises from the limited read-out resolution of the gravimeter and the large vibrational background noise.

5.6 Environmental effects

Environmental effects have a substantial impact on absolute gravity measurements and play a somewhat special role. Whether the resulting gravity offsets are considered systematic errors or valid measurements of environmental conditions is somewhat arbitrary. By convention, uncertainties arising from environmental signals are generally not included in the uncertainty estimate for any absolute gravimeter.

Apart from these fundamental considerations, for purely practical reasons it is important to understand the various environmental effects. Otherwise, it would not be possible to compare gravity measurements taken at different times and locations. Gravity variations resulting from environmental effects could also mask those caused by instrumental flaws. Table 4 summarizes some of the most important effects [15].

Table 4. Environmental effects.

Source	Magnitude/ μGal	Uncertainty/ μGal	Timescale
Solid Earth tides	300	0.2 to 0.5	Diurnal
Ocean loading	20	0.2	Diurnal
Air pressure	8	1 to 5	Hours-diurnal
Water table	Site-dependent	Site-dependent	Seasonal
Polar motion	10	< 0.01	12, 14 months

5.6.1 Tidal effects

Gravity tides, caused by the relative motion of Earth, Moon and Sun [54], are by far the biggest environmental effect ($\sim\pm 150$ μ Gal). We have already discussed tides, including the complications caused by ocean-loading effects [35-38, 55], in Section 4.3.

5.6.2 Atmospheric pressure

Variations in atmospheric pressure also affect local gravity. Higher air pressure corresponds to an increased density of air over the measurement, resulting in an upward-directed gravitational force and a lower total gravity value. Loading effects, i.e. pressure-induced deformations of the continental crust and the ocean surface, can cause additional changes [56]. The combined size of these effects is approximately 0.3 μGal/mbar and typical weather-induced pressure variations will thus change local gravity by several microgals. This makes it necessary to keep detailed pressure records in order to apply the appropriate corrections. Absolute gravity values are usually referenced to the nominal pressure at the measurement site, following DIN standard #5450 and using the formula [57]

$$p_{\mathrm{n}} = 1013.25 \left(1 - \frac{0.0065\, h_{\mathrm{m}}}{288.15}\right)^{5.2559}, \qquad (51)$$

where p_{n} is the nominal pressure in millibars and h_{m} the topographic elevation in metres.

5.6.3 Water table

Gravitational attraction from a fluctuating water table can easily cause gravity changes of several microgals. These variations are very site-dependent and are generally hard to predict. If the water table is modelled as an infinite slab then the effect is 42 μGal/m, scaled by the bulk porosity. The bulk porosity can vary from $< 3\,\%$ in competent granite to $> 50\,\%$ in sands [15]. Based on an annual rainfall of about 0.5 m, the table should vary by less than 0.05 m such that the variation on g should be less than 2 μGal.

5.6.4 Polar motion

Polar motion refers to the small movements of the Earth's rotational axis relative to its body. The corresponding variations in the distance of a measurement site from the rotational axis affect the centrifugal component of g. The resulting changes are typically a few microgals and have a characteristic time scale of one year. They can be computed precisely [57, 58], since the pole positions are known with exquisite accuracy from Very Long Baseline Interferometry measurements of quasar positions [36].

5.6.5 Geological processes

Several long-term geological processes can also affect local gravity readings. The most important ones are postglacial rebound [59] in regions previously covered by glaciers, and tectonic uplift. They can cause changes in elevation of up to several millimetres per year, corresponding to microgal-level changes in the local gravity value. Scandinavia and the Canadian shield of North America are exemplary sites showing postglacial rebound. The Andes mountain range is an example of a region showing substantial tectonic uplift. At our measurement site in California both processes should be negligible.

5.6.6 Local mass distribution

Local mass distributions, especially when they change over time, become important when gravity is measured at the microgal level. Most absolute gravity measurements taken for geophysical research are therefore conducted at sites specifically selected to control these influences.

A good site is removed from human activity. We, on the other hand, performed our measurement inside the Stanford physics building, during a time when we were surrounded by massive construction and demolition of neighbouring buildings.

Table 5 lists several local objects whose gravitational fields could affect our experiment. The more distant objects simply modify the absolute gravity value at the measurement site. The closest objects could also cause additional problems by producing non-linear gravity gradients (Section 5.5.2).

The nearby construction activities are currently our biggest concern. We have made substantial progress in eliminating systematic offsets in our own instrument since the comparison measurement with the falling corner-cube gravimeter (Section 4.4). The only classical reference value we have available to compare with our new atom interferometric gravity measurements is therefore several months old and may no longer be valid because of the construction activities.

5.6.7 Motion of building

Another problem with the location of our instruments, on the second floor of the physics building, is possible movements of the building and the resulting change in the measurement height.

One possible cause of building movements is temperature change. We can make a rough worst-case estimate for the effects of pure thermal expansion. Assuming that the floor of our laboratory is about $10\,\mathrm{m}$ above the foundations of the building, daily temperature variations of 10 K, and using the thermal expansion coefficient of $1.4 \times 10^{-5}/\mathrm{K}$ for concrete, we find a vertical displacement of $1.4\,\mathrm{mm}$. The corresponding change in gravity of 0.4 μGal would be difficult for us to detect. However, the value is large enough to raise concerns about additional, possibly greater, vertical motions caused by thermal deformations of the building.

The building might also move for a variety of other reasons, for example ground motions related to a nearby construction site or changes in the mechanical properties of the soil after rainfall. However, these effects would be difficult to identify and so far we have seen no clear evidence for their existence.

5.7 General tests and observations

We have also performed a variety of more general tests in the hope of uncovering additional problems.

5.7.1 Varying the interferometer pulse spacing

One possibility is to check if the measurement results are consistent for different spacings of the interferometer pulses. Figure 41 shows the result of such a test for the Doppler-sensitive interferometer configuration. Ideally, the measured gravity value

Table 5. Gravitational effects of various nearby objects.

Object	Mass/kg	Distance/m	Gravity/μGal	Gradient/ μGal m^{-1}	Angle/ degree	Gravity change/ μGal
Earth	6.0×10^{24}	6.4×10^{6}	9.8×10^{8}	308	0	9.8×10^{8}
Optical table	1000	1.5	3.0	4	0	3.0
Aluminium spacers	1	0.1	0.7	13	0	0.7
Experimental physicist (A. P.)	90	1.0	0.7	1.2	45	0.5
Loaded truck	40 000	10	2.7	0.5	45	2.0
Physics lecture hall (demolished)	2.0×10^{6}	50	5.0	0.2	90	0.0
Hole (excavated)	2.0×10^{7}	100	13.3	0.3	85	1.3

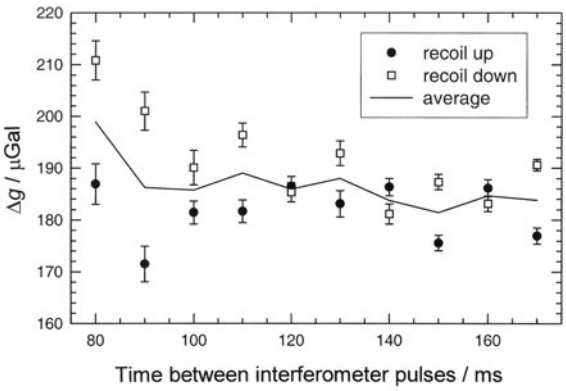

Figure 41. Effect of varying the interferometer pulse spacing. 1 μGal ≙ 1.0 mrad for $T = 80\,\mathrm{ms}$, 1 μGal ≙ 3.8 mrad for $T = 160\,\mathrm{ms}$.

should not depend on the pulse spacing, except for a very small contribution (6.3 μGal at $T = 160\,\mathrm{ms}$, 1.6 μGal at $T = 80\,\mathrm{ms}$) due to the change in the average height of the atoms in the presence of the local gravity gradient. Within the error bars this seems to be the case, but only if the effects of non-linear rf phase shifts are removed by averaging measurements for two directions of the photon recoil (Section 5.3.1).

5.7.2 Varying the time of the central interferometer pulse

We have also kept the pulse separation constant, but varied the time of the central π pulse. This changes according to the position of the atoms along their fountain trajectories during the interferometer sequence, and consequently any effect related to spatial position should manifest itself in the data. The method is also sensitive to any form of synchronous noise as it changes the timing relationship between the interferometer sequence and the rest of the experiment.

This measurement is complicated by rf phase shifts, which are a function of the time-dependent Raman difference frequency. Our set-up has two phase-locks that determine the Raman difference frequency (Section 3.2.2). To eliminate the rf phase-shift effects we can adjust the reference frequency f_{SRS} to accommodate the changes in the pulse time, while keeping the sweep of the other reference frequency f_{TEK}, which is responsible for most of the rf phase-shift effect, untouched.

The results for three different T are shown in Figure 42. The theoretical fit takes into account the parabolic dependence due to gravity gradients and the linear terms due to the finite speed of light and a changing k vector. The residual of about 1 μGal allows us to set that as an upper limit to a systematic effect as a result of any "trajectory effect".

Another cause of concern is the occasional rapid changes in the measured gravity value during long-term measurements. Figure 43b shows a few examples of such rapid changes during the comparison measurement

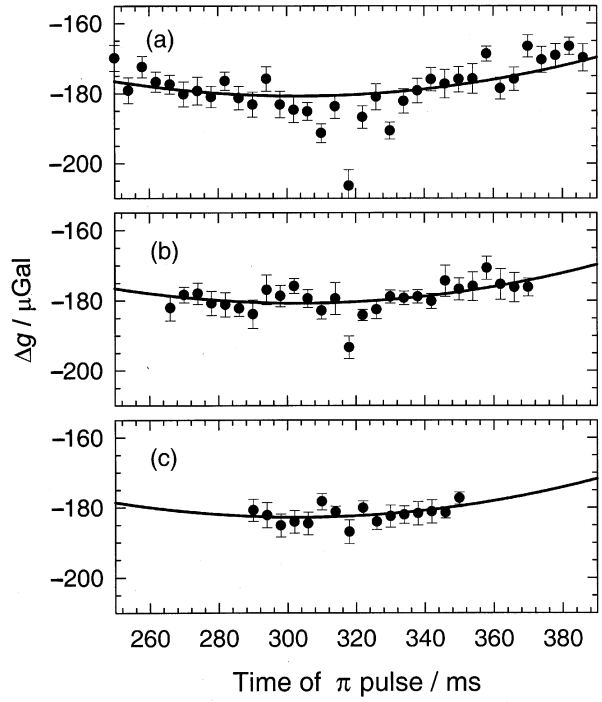

Figure 42. Effect of varying the time of the central interferometer pulse along the trajectory for (a) $T = 100$ ms, (b) $T = 120$ ms and (c) $T = 140$ ms. The data point at $t_\pi = 318$ ms was not considered for the fit as Doppler-free transitions are also driven causing substantial shift in measured g.

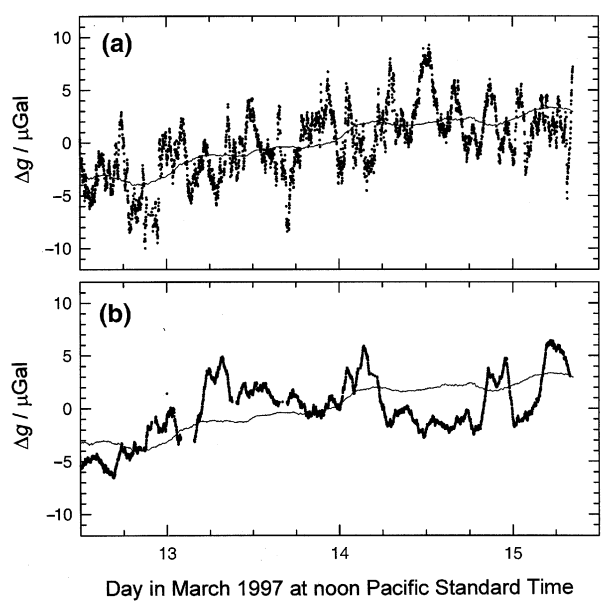

Figure 43. Rapid fluctuations in long-term gravity records in (a) FG5 gravimeter and (b) atom interferometer.

with the falling corner-cube gravimeter. It would be interesting to know if the effect is also present in the corner-cube data, but there it was completely masked by the higher levels of measurement noise as in Figure 43a.

The changes have typical rise and fall times of less than an hour, which should be too rapid for

most environmental effects such as tides or changes in air pressure. On the other hand, the data show a certain periodicity which suggests that tidal effects, most probably due to ocean loading, might play a role. We can also rule out changes in air pressure as the cause, since an independent recording of the pressure during the experiment shows no equivalent jumps.

Since the changes appear to be random, we consider them to be long-term noise of about 3 μGal and to produce no significant systematic shift provided that measurements are taken over a period of a few days.

5.8 Other effects

We have considered several additional systematic effects [23], but found that they are too small to be relevant for our experiment. We briefly list them here:

- General relativistic or higher-order special relativistic effects taken alone are smaller than $10^{-13}\,g$ as the atomic velocities and free-fall distances in our atomic fountain are relatively small.
- A Berry phase is not important as we do not use adiabatic processes to implement the beam splitters of the atom interferometer and the external electromagnetic field is too weak for significant Aharanov-Cashier effects.
- Transient phase shifts caused by the circuitry for pulsing the Raman beams cause gravity offsets of less than $10^{-13}\,g$ as we use relatively long ($>$ 40 μs) square pulses. However, they might become relevant for either very short ($<$ 1 μs) or shaped (Blackman, Hanning, ...) pulses.
- The ac Stark shifts caused by Raman leakage light should be less than $10^{-10}\,g$. Shifts caused by other near-resonant light are minimized by enclosing the laser system in light-tight boxes and using mechanical shutters for all the relevant beams.
- Changes in the refractive index of a cloud of cold atoms lead to gravity offsets of less than $10^{-12}\,g$.

5.9 Summary

Table 6 lists the systematic errors discussed in this section.

Recognizing and controlling the Coriolis sensitivity led to the greatest improvement in experimental accuracy, not counting the use of the correct value of the caesium D_2 wavelength. The next greatest improvement can be attributed to enclosing the active vibration-isolation system with magnetic shielding.

Currently, instrumental accuracy is mainly limited by the difficulty of minimizing the sensitivity of the interferometer to Coriolis forces, and by non-linear phase shifts in the rf electronics controlling the frequency difference of the Raman beams.

Table 6. The major sources of potential systematic effects are tabulated. Systematic effects that are less than 1 part in 10^{10} are not listed. These include timing, switching and chirping of pulses, Zeeman and dc Stark shifts, dispersion of light in air, Cs vapour and windows, tilt of Raman beam with respect to the vertical, standing-wave effect, leakage light, wavefront curvature and speckle, Berry phase, second-order Doppler shift, gravitational red shift and other relativistic effects. Environmental effects, while not part of our instrumental uncertainty, are important for comparing g measured at different times. Other environmental effects include water-table correction.

Systematic errors	$10^9 \times$ Relative uncertainty
Instrumental	
rf phase shift	2.0
Coriolis effect	2.0
ac Stark shift	1.0
Synchronous noise	1.0
Dependence on pulse timing	1.0
Retroreflection	0.6
Laser-lock offset	0.4
Cs wavelength	0.3
Gravity gradient	0.2
Cold collision	0.2
Synchronous vibration	0.2
Synchronous fields	0.2
Changing $\boldsymbol{k}$ vector	0.1
Overall instrumental uncertainty	**3.4**
Environment	
Pressure correction	1.0
Ocean loading	1.0
Other environmental effects	2.0

6. Conclusions

6.1 Current status

After implementing the improvements discussed in the last section we performed another atom interferometer measurement of g. Table 7 shows the results, including some of the corrections that we apply to the raw gravity value. While some of these corrections (polar motion, transfer to top of fountain) are simply a matter of convention, others (rf phase shift, caesium lock offset) are necessary to account for known systematic effects. The table also indicates the most important known measurement uncertainties and how they affect the final result.

Table 7. Calculation of atom interferometer gravity value.

	Value/μGal
Measured g value (tide-corrected)	979 933 179
Polar motion	−4.8
rf phase shift	−6.0
Caesium lock offset	−3.0
Finite speed of light	+0.3
Transfer to top of fountain	−6.0
Pressure correction	+0.7
Atom interferometer gravity value	979 933 158

We can compare this new result to the falling corner-cube measurement taken several months earlier, but we have to consider the possibility that the gravity value could have changed in the meantime. For example, changes in the local mass distribution caused by major construction work in the vicinity of our laboratory (Section 5.6.6) or seasonal changes in the water table (Section 5.6.3) could be responsible for differences of up to a few microgals.

Table 8 nevertheless shows that after including the measurement-height correction (Section 5.5.2) our value for g and that measured by the classical absolute gravimeter now agree within $(7 \pm 7) \times 10^{-9}\, g$. This is consistent with the estimated systematic uncertainty and is by far the highest accuracy achieved by any matter-wave interferometer measurement to date. It is also in marked contrast to the consistent discrepancies of order $1\,\%$ to $10\,\%$ between classical and neutron interferometric gravity measurements [60-62, 50].

Table 8. Comparison between atom interferometer and falling corner-cube gravity measurements. An independent measurement of the gravity gradient (using LaCoste-Romberg spring-type gravimeters) is used to compensate for the difference in measurement height. The atom interferometer gravity value is slightly higher.

	Value/μGal
Atom interferometer gravity value	979 933 158 ± 3
Falling corner-cube gravity value	– 979 933 298 ± 2
Measurement-height correction	+ 147 ± 5
Environmental effects	0 ± 2
Difference	7 ± 7

6.2 Future prospects

To improve the measurement accuracy even further it will be necessary to improve control of the Raman phase and to address problems arising from Coriolis forces and environmental effects. A good way to address these issues would be to build portable versions of the atom interferometer in order to facilitate measurements and comparison studies with different instruments at sites more suitable than our current laboratory. There are several other possible improvements. Horizontal velocity selection can reduce Coriolis effects. Additional Raman pulses can increase the separation between interferometer paths and, consequently, the measurement sensitivity. Alternatively, one can use a different intermediate state and shorter-wavelength Raman lasers to achieve the same effect. Using a taller fountain and a pulse spacing larger than the current $T = 160\,\mathrm{ms}$ can decrease the relative importance of Raman phase errors. Keeping the atoms in the same internal state for both interferometer paths would substantially reduce the sensitivity to magnetic and electric fields.

For practical applications, many of which only require an accuracy of $\Delta g/g \sim 10^{-8}$, one might also try to design a smaller, simplified instrument. In this case, it might be possible to use either continuous atomic beams or high repetition rates ($> 10\,\mathrm{Hz}$) to circumvent aliasing problems and alleviate the need for a complex active vibration-isolation system.

Future experiments could use the techniques developed here to perform tests of the equivalence principle or to measure Newton's gravitational constant, G. It should also be possible to exploit the wavelength dependence of the interferometer phase shift. Using different intermediate states for the Raman transitions would allow the wavelengths of all the optical transitions that are accessible in this way to be compared with a relative precision of 10^{-10} or better.

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